

WALLIS FORMULAE: The formulae

$$\int_0^{\pi/2} \cos^{2n+1}(t) dt = \frac{2, 4, 6, \dots, 2n}{1, 3, 5, \dots, (2n+1)} \text{ and}$$

$$\int_0^{\pi/2} \cos^{2n}(t) dt = \frac{1, 3, 5, \dots, (2n-1)}{2, 4, 6, \dots, 2n} \cdot \frac{\pi}{2}.$$

Wallis' product for pi is computed from these formulae by repeated integration by parts. It is defined as the infinite product $\frac{\pi}{2} = \frac{2 \times 2 \times 4 \times 4 \times 6 \times 6 \dots}{1 \times 3 \times 3 \times 5 \times 5 \times 7 \times \dots}$. It was named after

English mathematician, John Wallis (1616-1703) who was a major contributor to the development of infinitesimal calculus.

WALSH DIAGRAM (ANGULAR COORDINATE DIAGRAM; CORRELATION DIAGRAM): A diagram that relates the energy levels of separate atoms to the energy levels of diatomic molecules and united atoms, the two limiting states being correlated with each other. It explains why molecules are more stable in certain spatial configurations. It is used for making predictions about the geometries of small molecules.

WALTON, ERNEST THOMAS SINTON (1903-1995): Irish physicist, educated at Methodist College (Belfast), Trinity College (Dublin) and Cambridge, who became a fellow of Trinity College in 1934 and professor of natural and experimental philosophy there in 1946. He shared the 1951 Nobel Prize in Physics with Sir John Cockcroft for their pioneering work in transmuting atomic nuclei by bombarding elements with artificially accelerated atomic particles.

WAN (WIDE AREA NETWORK): A network of computers such as the Internet located at geographically separate sites. Many wide area networks span long distances via telephone lines, fibre-optic cables or satellite links. They can also be composed of smaller LANs that are interconnected. For home routers the port the router connects to the Internet connection is often labelled as a WAN, Network or Internet port, since it is what allows home network to communicate with the Internet network.

WANING: Pertaining to the Moon, it is the period during which its illuminated surface that is visible decreases in size.

WAP (WIRELESS APPLICATION PROTOCOL): A specifications for a set of communication protocols that enables a user to browse online services such as e-mail, chat, information relating to the weather, traffic conditions and shopping via a special type of mobile phone. It was started in the 1990s by Ericsson, Motorola, Nokia and Unwired Planet (now known as Openwave) in June 1997 and now includes hundreds of member companies that are infrastructure providers, software companies and content providers. WAP is the mobile phone equivalent of the World Wide Web. Newer mobile phones include WAP browser software to allow users access to WAP sites.

WARBLE FLY: A parasitic fly, whose larvae infest cattle. The warble flies *Hypoderma lineatum* and *Hypoderma bovis*, also called cattle grubs or heel flies, are large, heavy and beetle-like. They deposit their eggs on the legs of cattle. The larvae penetrate the skin, migrate through the body and produce a lump or warble, on the animal's back. Mature grubs emerge and drop to the ground to pupate.

WARBURG EFFECT: The inhibition of carbon dioxide assimilation and photosynthesis by atmospheric oxygen, described by O. Warburg in 1920. This phenomenon was later discovered to be due to photorespiration.

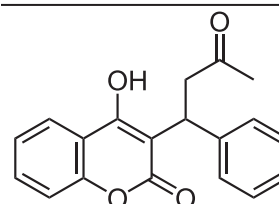
WARE GROWERS (WARE POTATOES): Farmers who grow potatoes for consumption, not for seed. Ware potatoes are those potatoes that are grown for human consumption as fresh or processed products. Ware or eating, potatoes are a traditional crop in Northern Ireland and are of potential interest to many organic producers because of the following factors: (i) there is good interest from consumers and

producing quality crops can be profitable; (ii) potatoes provide a break crop from grass/clover leys in the rotation; (iii) cultivations for potatoes provide opportunities to control weeds; (iv) organic potato production can fit into a planned rotation on many organic units.

WAREZ: In computer science, it is copyrighted software that is pirated by removing its protection codes de-activated and downloaded by using File Transfer Protocol (FTP).

WARFARIN [3-(ALPHA-ACETONYL-BENZYL)-4-HYDROXYCOUMARIN]: A

synthetic anticoagulant drug, with the chemical formula $C_{19}H_{16}O_4$ and marketed as Coumadin. Warfarin interferes with the liver's metabolism of vitamin K, leading to production of defective coagulation factors. It is used therapeutically in clinical medicine and to poison rats. Many rats in some areas are now resistant to the drug.



Chemical Structure
of Warfarin

WARING'S PROBLEM: In mathematics, it is the conjecture that every natural number can be written as the sum of a fixed smallest number, $g(n)$, of n^{th} powers of integers. Waring originally speculated that $g(2) = 4$, $g(3) = 9$ and $g(4) = 19$. Lagrange's theorem shows that $g(2) = 4$. The exact value of $g(n)$ is known for all but finitely many n ; for $n < 20\,000$, $g(n)$ is the integer part of $(3/2)^k + 2^k - 2$.

WARM BOOT: The process of restarting a computer once it has been turned on, in contrast to a cold or hard boot, which refers to turning on the computer after it has been powered off by pressing the power button on the computer.

WARM FRONT: A zone of transition between warm and cold air masses when warm air advances, displacing cooler air, which may bring steady rain, sleet, snow or thunderstorms. Warm fronts typically move from southwest to northeast and the air behind a warm front is warmer and more moist than the air ahead of it. On weather maps, a warm front is represented with a red line of semicircles pointing in the direction of travel.

WARM-BLOODED (HOMOEOTHERMIC): A term for describing an animal that maintains its internal body temperature at a relatively constant value independent of environmental temperature by using metabolic processes to counteract fluctuations in the temperature of the environment. Examples of warm-blooded organisms are mammals and birds (higher vertebrates).

WARNING COLOURATION (APOSEMATIC COLOURATION): The conspicuous markings of an animal such as the yellow and black striped abdomen of the wasp, which gives a warning of its sting. This makes it easily recognised and warn would-be predators that it is a poisonous, foul-tasting or dangerous species. Studies have shown that all vertebrates including insectivorous birds associate greens and blues with safety and inherently regard red orange, yellow and white as signs of danger. It is also widely accepted that patterns incorporating stripes or spots draw attention to objects. The signals are beneficial for both the predator and prey, who both avoid potential harm.

WARP: The uncontrolled torsional change of shape or outline of a material due to change in moisture content or to exposure to varying temperatures.

WART: A small harmless rough lump that grows, usually, on the feet, hands or genitals, usually caused by a papillomavirus, which triggers overproduction of epidermal cells. It is also a growth on a plant that looks like a wart.

WASH BOTTLE: A bottle with a nozzle, whose contents can be squeezed to rinse, sterilise, and dry various pieces of laboratory glassware, such as test tubes and round bottom flasks. For safety reasons, wash bottles are labelled with various colour codes and markings for recognition of contents.

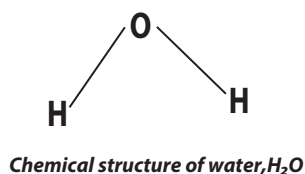
WASH OUT: The removal from the atmosphere of gases and sometimes particles by their solution in or attachment to raindrops as they fall.

WASHING SODA (SODA ASH): A crystalline form of sodium carbonate with the chemical formula Na_2CO_3 , which readily effloresces to form a white powder, the monohydrate. It has a cooling alkaline taste and can be extracted from the ashes of many plants. It is synthetically produced in large quantities from table salt in a process known as the Solvay process. Its uses include softening water, manufacture of glass, as an electrolyte, etc.

WASTE (WASTE PRODUCT): 1. An unwanted or unusable materials given off by a process and are discarded. Examples are household waste and industrial waste. Some components of waste have economical value and can be recycled once correctly recovered. 2. Any product of metabolism that is not required for further metabolic processes and excreted from the body. Examples include nitrogenous compounds, such as urea and ammonia, carbon dioxide and bile. Organic waste can be treated and reused as fertiliser. 3. Any radioactive waste.

WASTE DISPOSAL: A removal of or getting rid of waste such as contaminated objects. The methods of waste disposal vary according to the materials in the waste and include incineration, burial at designated sites and dumping at sea. Organic waste can be treated and reused as fertiliser.

WATER: A colourless, tasteless and odourless chemical substance with the chemical formula, H_2O that freezes at 0°C (273 K) and boils at 100°C (273 K) at standard pressure, relative density is 1.0 at 4°C (277 K). It is the best known solvent and functions as a solvent for a wide variety of chemical substances and facilitates industrial cooling and transportation. Its molecule contains one oxygen and two hydrogen atoms connected by covalent bonds. Water covers 70.9% of the Earth's surface and is vital for all known forms of life.



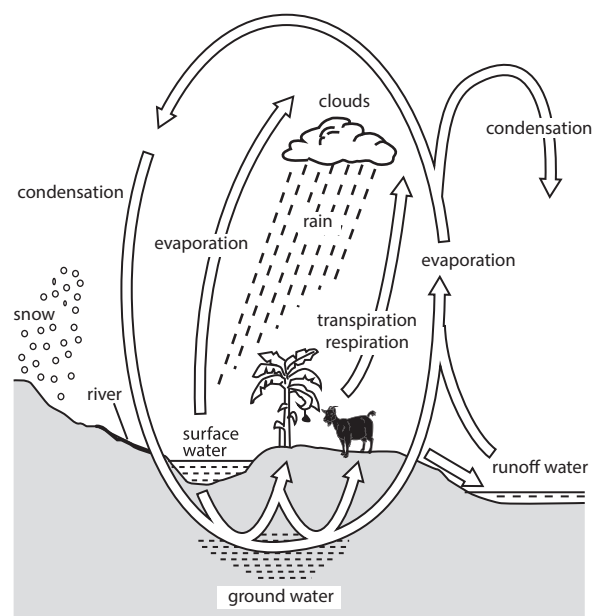
WATER ABSTRACTION (WATER EXTRACTION; GROUNDWATER ABSTRACTION): The process of removing or taking water from any surface or underground source for some purpose such as irrigation or treatment to produce drinking water.

WATER BALANCE: 1. A state in which the water lost in an area by evaporation or by runoff is replaced by water received in the form of rain. 2. The balance between intake and excretion of fluids in which the water lost by the body in urine and perspiration or by other physiological processes is balanced by water absorbed from food and drink. The kidneys control the volume of extracellular water by excreting water. Ingestion of sodium chloride (salt) raises the osmotic pressure of the extracellular water, causing thirst.

WATER CULTURE (HYDROPONICS): The growth of plants in a sterile medium, such as sand or vermiculite, to which nutrients are added in a balanced liquid fertiliser.

WATER CYCLE: The continuous circulation of water between the atmosphere and the land and oceans. Rain plays a key role in

the hydrologic or water cycle in which moisture from the oceans evaporates, condenses into clouds, precipitates back to Earth and eventually returns to the ocean via interception, infiltration, subterranean percolation, overland flow, runoff and other complex processes into streams and rivers to begin the cycle again.



The water cycle

WATER DISPLACEMENT: A case of fluid displacement in which an object is put into water and the water level rises. This happens because the object takes up space and the water is forced to go up.

WATER GAS: A toxic mixture of carbon monoxide (50%), hydrogen (40%) and small amounts of methane, carbon dioxide and nitrogen, made by passing steam and air over heated coke or coal. It is used as fuel for heating, lighting and power.

WATER GLASS (SODIUM SILICATE; SOLUBLE GLASS; LIQUID GLASS): Viscous colloidal solution of sodium silicates in water with the chemical formula $\text{Na}_2\text{O}_3\text{Si}$, density 2.4 g cm^{-3} and melting point 1088°C (1361 K). It is a compound containing sodium oxide (Na_2O) and silica (silicon dioxide, SiO_2) that forms a glassy solid with the very useful property of being soluble in water. It is a transparent, glassy or crystalline solid. It is used as a convenient source of sodium for many industrial products, as a builder in laundry detergents, as a binder and adhesive, as a flocculant in water-treatment plants, as a preservative for eggs, used in plaster and cement and in various purification and refining processes.

WATER HARDNESS (HARD WATER): Water that contains certain salts, such as those of calcium or magnesium, which forms insoluble deposits in boilers and forms precipitates with soap. Water's hardness is determined by the concentration of multivalent cations (positively charged metal complexes) in the water. Multivalent cations are cations with a charge greater than $1+$, usually $2+$. Common cations found in hard water include calcium cations (Ca^{2+}) and magnesium cations (Mg^{2+}). These ions enter a water supply by leaching from minerals within an aquifer. Common calcium-containing minerals are calcite and gypsum. There are two types of hard water: temporal and permanent. **Temporal hardness** is caused by the presence of dissolved carbonate minerals (calcium carbonate and magnesium carbonate). When dissolved, these minerals yield calcium and magnesium cations (Ca^{2+} , Mg^{2+}) and carbonate and bicarbonate anions (CO_3^{2-} , HCO_3^-). This

hardness can be reduced either by boiling the water or by the addition of lime (calcium hydroxide) through the process of lime softening. **Permanent hardness** is usually caused by the presence of calcium and magnesium sulfates and/or chlorides in the water, which become more soluble as the temperature increases. It cannot be removed by boiling but can be easily removed using a water softener or ion exchange column.

WATER HOLDING CAPACITY (SOIL WATER HOLDING CAPACITY; WATER-HOLDING CAPACITY): The amount of water that a given soil can hold after gravitational water loss has ceased. This determines the **available soil water holding capacity**, which is the amount of water volume between field capacity (the maximum amount of water the soil can hold) and wilting point (where the plant can no longer extract water from the soil).

WATER MELON: A vine (*Citrullus lanatus*) of the family Cucurbitaceae; native to Africa and cultivated for its large edible fruit, also known as water melon, having a hard green rind and sweet watery pink or reddish flesh. It is very low in saturated fat, cholesterol and sodium and also a good source of potassium and Vitamins A and C. However, a large portion of its calories come from sugars.

WATER MOULDS (SAPROLEGNIALES): An order of the Oomycetes containing some 32 genera and about 200 species of aquatic, mainly saprobic, fungi (e.g. Saprolegnia), often with diplanetic zoospores. Some are parasitic on fish, for example, *Saprolegnia parasitica*, which often attacks salmon and goldfish.

WATER OF CRYSTALLISATION: Water, which is chemically bonded with crystals or water molecules incorporated in a crystalline substance that are typically necessary for its properties and structure but capable of being removed by sufficient heat.

WATER POLLUTION: A condition caused by any addition to fresh or seawater by materials such as sewage effluent, chemicals, detergents and fertiliser runoff that disrupts biological processes or causes a health hazard. Common pollutants include nitrates, pesticides and sewage.

WATER POTENTIAL: The physical property that predicts the direction in which water molecules will flow across a membrane governed by solute concentration and applied pressure. It is denoted by ψ and measured in kilopascal (kPa). It can be expressed mathematically as: $\psi = \psi_s + \psi_p$, where ψ is the water potential, ψ_s is the solute potential and ψ_p is the pressure potential. Water tends to move from a region of high water potential to a region of low water potential. Water potential is important in describing osmosis in cells and extracellular systems. Osmosis in plant is now described in terms of water potential. In soils and other extracellular systems, another factor called **matrix potential** can contribute significantly to water potential. The water potential of pure water is zero.

WATER PURIFICATION: The treatment applied to water such as removal of scale-forming calcium and magnesium ions from hard water or replacing them by the more soluble sodium ions in order to make it safe to drink. It also includes filtering water through sand and gravel to remove large particles and then adding a disinfectant, for example, chlorine to kill harmful microorganisms.

WATER RETENTION (FLUID RETENTION): A condition in which excess water is retained in the circulatory system or within the tissues or cavities of the body. This is caused by an injury or disabilities including kidney diseases, cirrhosis of the liver and hypertension.

WATER RETENTION CURVE (MOISTURE-RETENTION CURVE): A graph showing the water content of soil and the applied tension or pressure. It gives the relation between soil moisture suction and soil moisture content. Points on the graph are usually obtained by increasing or decreasing the applied tension or pressure over a specified range.

WATER SOFTENER: Any substance or unit that reduces the hardness from water by removing calcium and magnesium. This prevents the formation of insoluble scale in pipes and tanks and does not form a precipitate with soaps or interfere with other cleaners. Water softeners usually consist of zeolite or an ion-exchange resin in a tank connected directly into the water system. The zeolite or resin contains sodium ions that change places with the calcium and magnesium ions dissolved in the water. When the zeolite or resin becomes exhausted (when most of its exchangeable sodium is replaced with calcium and magnesium), it can be regenerated by washing with a strong solution of common salt, which removes the calcium and magnesium and replaces them once again by sodium. Indispensable in many industries, water-softening units are also used in homes.

WATER STRESS: A situation in which the demand for water exceeds the available amount during a certain period or when poor quality restricts its use. A water stress situation arises if consumption exceeds 20% of renewable water supply.

WATER STRIDER (POND SKATER): Any of various predator insects of the family Gerridae, having slender long legs to walk on water and feeds on dead insects.

WATER TABLE: The upper level of ground water, below which the soil or rocks are permanently saturated with water. The water table separates the groundwater zone (zone of saturation) that lies below it from the zone of aeration that lies above it. It fluctuates both with the seasons and from year to year because it is affected by climatic variations and by the amount of precipitation used by vegetation.

WATER VAPOUR (AQUEOUS VAPOUR): The gaseous phase of water that is invisible. It is one state of water within the hydrosphere. Water vapour can be produced from the evaporation or boiling of liquid water or from the sublimation of ice.

WATER VASCULAR SYSTEM: A network of hydraulic canals unique to echinoderms such as sea stars and sea urchins that branches into extensions called tube feet and used in locomotion, feeding and gas exchange.

WATER-BORNE: Travelling on, transported or transmitted by water.

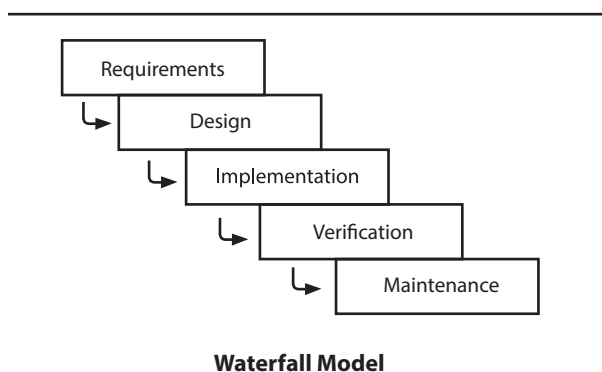
WATER-BORNE DISEASE: A disease transmitted by drinking water or by contact with potable or bathing water. Examples are diarrhoea, guinea-worm and cholera.

WATER-USE EFFICIENCY: Carbon gain in photosynthesis per unit water lost in evapotranspiration. It can be expressed on a short-term basis as the ratio of photosynthetic carbon gain per unit transpirational water loss, or on a seasonal basis as the ratio of net primary production or agricultural yield to the amount of available water.

WATER-STABLE AGGREGATE (WSA): A soil aggregate that is stable to the action of water, such as falling drops. This is an indication of a soils' resistance to compaction.

WATERFALL: Water that drops abruptly and nearly vertically. It occurs when a river flows over a bed of rock that resists erosion or when weaker rock downstream are worn away, creating a steep vertical drop and plunge pool into which the water falls. Waterfalls of small height and less steepness or a series of small falls are called **cascades**. Stretches of river that flows swiftly and turbulently through shallows and narrows or past boulders or other obstacles are called **rapids**. Some famous waterfalls are Victoria Falls, the largest waterfall in the world, more than 1 kilometre long, located on the Zambezi River on the border of Zimbabwe and Zambia; and Niagara Falls in North America. Waterfalls generate kinetic energy, which can be converted into hydroelectricity by a generator. An example is the Niagara Falls, first diverted for hydroelectricity in 1893.

WATERFALL MODEL: A sequential design method used during software development in which progress is seen as a steady flow downward as seen in a waterfall.



WATERING CAN: A portable metal or plastic container with a long spout and a detachable perforated cap, used for watering plants.

WATERLOGGED: 1. Filled or saturated with water so as to become spongy or marshy and unmanageable. 2. Of wood, so saturated with water, that it barely floats. 3. Of a ship, so full of water that it can barely float and has become so heavy as not to be manageable by the helm.

WATERMARK: 1. A transparent mark or text superimposed on paper such as a banknote to distinguish it from a counterfeit or an illegally produced one, or to make it harder to reproduce illegally. This mark can be seen when the paper is held to the light. 2.

Digital watermark or **forensic watermark** is data embedded into intellectual property (IP) that is used to identify the source of illegal copies. Digital watermarks provide copyright protection to digital intellectual property (IP), such as audio, video or image data. Digital watermarks are invisible, or in the case of audio clips, inaudible but serve as signals when copyrighted materials are downloaded or reproduced. 3. **High Water mark (High Water-mark)** is a mark showing the highest level to which water has risen over a period. It is also a line showing the highest level of high tide.

WATERSHED: The area of land that is drained by a river system

WATSON-CRICK BASE PAIRS: The type of hydrogen bonded base pairs found in DNA or comparable base pairs found in RNA. The base pairs are adenine-thymine (A-T) and guanine-cytosine (G-C) in DNA and adenine-uracil (A-U) and guanine-cytosine (G-C) in RNA.

WATSON-CRICK MODEL (DOUBLE HELIX): A three-dimensional model molecular structure of DNA, consisting of two polynucleotide strands wound in the form of a double helix and joined in a ladder-like fashion by hydrogen bonds between the purine and pyrimidine. It was co-discovered in 1953 by James Dewey Watson, (1928-), American molecular biologist, geneticist and zoologist and Francis Harry Compton Crick (1916-2004) English molecular biologist, biophysicist and neuroscientist.

WATT: The SI unit of power or work done per unit time, with the symbol W and defined as the power used when one joule of work is done in one second or the power produced by a current of one ampere acting across a potential difference of one volt. It is used as a measure of electrical and mechanical power. $1 \text{ W} = \text{J s}^{-1} = \text{m}^2 \text{ kg s}^{-3}$. It is named after James Watt (1736-1819), a Scottish engineer and inventor.

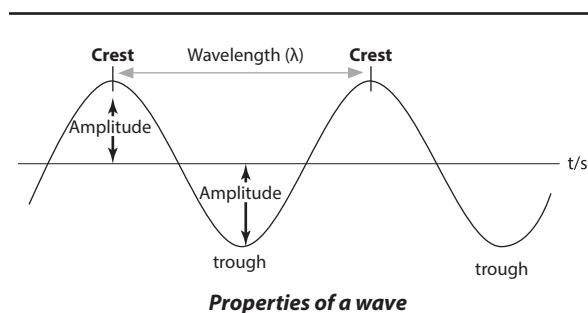
WATT, JAMES (1736-1819): Scottish engineer and inventor, who was self-educated. He was an instrument maker and later an engineer on the Forth and Clyde Canal. Watt's major improvement to Thomas Newcomen's steam engine was the use of a separate condenser (1769), which reduced the loss of latent heat and greatly increased its efficiency. With Matthew Boulton, he began to manufacture his new engine in 1775. In 1781 he added rotary motion to replace the up-and-down action of the original engine. In 1782 he patented the double-acting engine, in which the piston pushed as well as pulled.

This engine required a new method of rigidly connecting the piston to the beam, a problem he solved in 1784 with an arrangement of connected rods that guided the piston rod in a perpendicular motion. His application of the centrifugal governor for automatic control of the speed of the engine (1788) and his invention of a pressure gauge (1790) virtually completed the Watt engine, which had immense consequences for the Industrial Revolution. He introduced the concept of horse power, the **watt**, a unit of power, named after him.

WATTLE: 1. Flesh, often brightly coloured hanging down from the neck or throat of a bird, for example a turkey, cock and also some lizards. 2. A structure of sticks, reeds, branches or twigs woven over and under thicker upright sticks and used for fences, walls, etc.

WATTMETER: An instrument designed to measure the magnitude of the power in an alternating electric circuit. It may be scaled in watts, kilowatts or megawatts. In a direct-current circuit, power is usually determined by separate measurements of the voltage and the current.

WAVE: A periodic disturbance in a medium that travels through space and time, usually by the transfer of energy. A **mechanical wave** is a wave that propagates through a medium owing to restoring forces resulting from its deformation. For example, sound waves propagate via air molecules bumping into their neighbours. This transfers some energy to these neighbours, which will cause a cascade of collisions between neighbouring molecules. When air molecules collide with their neighbours, they also bounce away from them (restoring force). This keeps the molecules from actually travelling with the wave. The main characteristics of a wave include its speed of propagation, frequency, wavelength, amplitude, crest or peak and trough. The amplitude, which corresponds to the energy of the wave, is the distance from the baseline to a crest or the baseline to a trough and is the maximum difference of the distributed quantity from its mean value and the speed of propagation is the distance covered by the wave in unit time. The wavelength, denoted by λ is the distance in metres between successive points of equal phase disturbances (cycles) in unit time; crest is the highest point on a wave; and trough is the lowest point. The different types of waves include travelling or progressive waves, transverse waves, stationary waves, longitudinal waves, etc. In a **travelling wave (progressive wave)**, energy is transferred from one place to another by vibrating particles or fields, while a **stationary** or **standing wave** is a wave in which the profile of the wave does not move through the medium but remains stationary. A **stationary wave** results when a travelling wave is reflected back along its own path. In the case of **transverse waves**, the disturbances are at right angles to the direction of propagation. A **transverse wave** may be mechanical, such as the wave projected in a taut string that is subjected to a transverse vibration; or it may be electromagnetic, such as light, x-ray or radio waves. In **longitudinal waves**, the particles in the medium move parallel to the direction of propagation of the wave. Longitudinal waves have compressions (where the particles are closest to each other) and rarefactions (where the particles are furthest from each other) instead of crests and troughs. The vibrations of the particles are alternately compressed and rarefied by displacements in the direction of propagation. Longitudinal waves occur, for example, in sound waves, seismic P-waves as created by earthquakes and explosions, ultra sounds, vibrations in gas, and oscillations in spring.



WAVE EQUATION: A partial differential equation of the form: $\nabla^2 u = \left(\frac{1}{c^2}\right) \partial^2 u / \partial t^2$, where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is the Laplace operator. It represents the propagation of a wave, where u is the displacement and c the speed of propagation. If waves are emitted by a source per unit time and each occupies length λ , then a distance $f\lambda$ is swept out by the wave in each unit of time; this is the velocity, V . $V = f\lambda$. The general equation of a stationary wave is given by $Y = 2a \cos kx \sin \omega t$, where Y is the vertical displacement at any given time, a is the amplitude of the individual wave, ω is the angular velocity and K is the wave constant.

WAVE FILTER: An instrument that clarifies a signal by filtering out all other noise. It is useful for receiving communications in deep space, where there are other stellar phenomena that emit frequencies too. These extra frequencies become part of the white noise always present in space. A wave filter will keep a satellite from being confused by this extraneous noise.

WAVE FRONT: 1. The continuous line or surface within two or three dimensional mediums through which waves are passing or that joins points of the same phase in a wave travelling through a medium or a continuous line or surface including all the points in space reached by a wave or vibration at the same instant as it travels through a medium. It is the locus of all adjacent points at which the disturbances are in phase. 2. Lines drawn to represent a progressive wave. They usually show the places of maximum amplitude of the wave.

WAVE FUNCTION: The mathematical expression of a function $\Psi_{(x,y,z,t)}$ involving the coordinates of a particle in space. This function appears in Schrödinger's equation in quantum mechanics. For a particle in a given system such as an electron in an atom, depending on the boundary conditions, the solution is a set of allowed wave functions (eigen functions) of the particle, each corresponding to an allowed energy level. The physical significance of the wave function is that the square of its absolute value, $|\Psi|^2$ at a point is proportional to the probability of finding the particle in a small element of volume, $dx dy dz$, at that point.

WAVE GUIDE (WAVE-GUIDE): An elongated hollow tube through which microwave electromagnetic radiation can be transmitted with relatively little attenuation. The electromagnetic radiation is conducted in a reverse sense in that radiation travels through the insulator, usually the atmosphere, surrounded by metal along which the waves propagate by multiplying internal reflection. They often have a rectangular cross section, but some have a circular cross section. In transverse electric (TE) modes the electric vector of the field has no component in the direction of propagation. In transverse magnetic (TM) modes, the magnetic vectors have no such component.

WAVE MECHANICS: A formulation of quantum mechanics in which the dual wave-particle nature of such entities as electrons is described by the Schrödinger's equation. The modern theory of matter holds that elementary particles such as electrons, protons and neutrons have wavelike properties. Schrödinger showed that the wave mechanics is equivalent to matrix mechanics and also postulated a wave mechanics that bears the same relation to Newtonian mechanics.

WAVE MOTION (WAVE PROPAGATION; WAVE PROPAGATION): The process in which a fluctuation at one point is transmitted to another with no net transport of the propagating medium. Examples include the motion of electromagnetic waves, sound waves, hydrodynamic waves in liquids and vibration waves in solids.

WAVE NUMBER: The number of cycles of a wave in unit length or distance. It is the reciprocal of wavelength. Its symbol is k and the SI unit is the reciprocal metre (m^{-1}).

WAVE PACKET: The superpositions of waves with one most important wave number k , but with several other wave numbers near k . Wave packets are useful for the analysis of scattering and for describing localised particles of both matter and photons. The

motion of a wave packet is in line with the motion of an analogous classical particle, provided that the potential energy change across the dimensions of the packet is tiny. This proposal is called **Ehrenfest's theorem**, named after the Dutch physicist Paul Ehrenfest.

WAVE PARTICLE DUALITY: The proposition that waves carrying energy may have a corpuscular aspect and that particles may have a wave aspect. The more convenient between the two models will depend on the properties the model intends to explain. For example, waves of electromagnetic radiation need to be regarded as particles, called photons, to explain the photoelectric effect while electrons need to be visualised as de Broglie waves in electron diffraction.

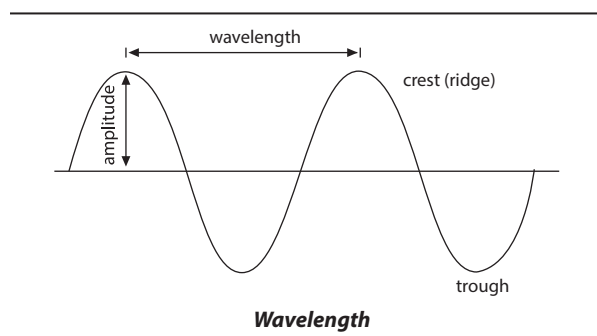
WAVE POWER: Energy produced for domestic or industrial use by harnessing and converting the energy of sea waves, for example, for electricity generation, water desalination or the pumping of water (into reservoirs). In electricity generation, a series of bobbing floats are anchored off shore and the energy produced by the motion of the floats is used to turn a generator.

WAVE THEORY: The theory that the behaviour of light or any other electromagnetic radiation can be explained by assuming that it travels in waves. It is a first step to a grand united nature theory that helps us to understand the laws of nature, to generate new ideas and to contribute to research. Wave theory explains and connects theories such as those of Einstein, Darwin and quantum mechanics with every formation of matter, even living matter and thought.

WAVE VECTOR: A vector that determines the magnitude of a wave and its direction. The magnitude is given by $2\pi k$, where k is the wave number or equal to $2\pi/\lambda$, where λ is the wavelength.

WAVEFORM: The shape of a wave or the pattern representing a vibration obtained by plotting the amplitude of the wave with respect to time. There are an infinite number of possible waveforms. They include: (i) the **square wave**, in which a quantity such as voltage alternately assumes two discrete values during repeating periods of time; (ii) the **sine wave**, having a form which, if plotted, would be the same as that of a trigonometric sine or cosine function; (iii) the **sawtooth wave**, which is a repeating waveform that rises from zero to maximum value, linearly drops back to zero and repeats; (iv) the **triangular wave**, the amplitude rises along a straight line to a given maximum, then returns quickly to the baseline again following a linear path. Alternatively, such a wave may rise quickly to a given peak, then drop gradually following a linear path. (v) The **arbitrary wave**, a recurrent waveform which takes on an arbitrary shape over one complete cycle; this shape is then repeated in successive cycles.

WAVELENGTH: The distance between successive particles vibrating in phase; or the distance between corresponding points in successive forms such as two successive crests or two successive troughs. In transverse waves, wavelength is measured from crest to crest or from trough to trough. In longitudinal waves, it is measured from compression to compression or from rarefaction to rarefaction. Wavelength, λ , is equal to the speed v of a wave in a medium divided by its frequency f or $\lambda = v/f$. The wavelength of light determines its colour, for example, red light has a wavelength of about 700 nanometres.



WAVELENGTH CONVERTER: It converts radiation at one wavelength to radiation at another detectable wavelength or at a wavelength of improved responsivity of the detector. The classical wavelength converter consists of a screen of luminescent material that absorbs radiation and radiates at a longer wavelength. Such materials are often used to convert ultraviolet to visible radiation for detection by conventional phototubes. In X-ray spectroscopy, a converter that emits optical radiation is called a **scintillator**. In most cases wavelength conversion is from short to long wavelength, but in the case of conversion of long to short wavelength the process is sometimes called **upconversion**. Wavelengths of coherent sources can be converted using nonlinear optical techniques. A typical example is frequency doubling.

WAVELENGTH DISPERSION: Spatial separation of characteristic X-rays according to their wavelengths.

WAVELENGTH DIVISION MULTIPLEXING (WDM): In communication (fibre optics), it is a technology that combines (multiplexes) a number of different wavelengths onto a single carrier.

WAVELENGTH ERROR: The error in absorbance which may occur if there is a difference between the (mean) wavelength of the radiation entering the sample cell and the indicated wavelength on the spectrometer scale.

WAVELENGTH-DISPERSIVE X-RAY FLUORESCENCE ANALYSIS: A kind of X-ray fluorescence analysis involving the measurement of the wavelength spectrum of the emitted radiation, for example, by using a diffraction grating or crystal.

WAVELENGTH-DIVISION MULTIPLEXING (WDM): In communications, it is a method of combining multiple optical carrier signals at various infrared (IR) wavelengths onto a single optical fibre for transmission along fibre optic media and then split apart with a demultiplexer. It enables bidirectional communications over one strand of fibre, as well as multiplication of signal capacity.

WAVELET: A function that provides succinct and accurate representation of time series and arrays of spatial data. When the data consists of, for example, images to be sent across computer networks, they have great potential as tools for data compression. Its representation as $g(t)$, a continuous function of time t has the form $g(t) = a\varphi(t) + b\psi(t) + \sum_{n,m} c_{n,m} \psi(2^n t - m)$, where m and n are integers

with $n \geq$ and $m \geq 0$. The function ψ is the mother wavelet and φ is the scaling function or father and this wavelet representation of $g(t)$ has similarities to a Fourier series. It is more flexible since both φ and ψ are chosen to take the value 0 outside time intervals. The

functions φ and ψ are related by the equations $\varphi(t) = \sum_{k=0}^N h_k \varphi(2t - k)$

and $\psi(t) = \sum_{k=0}^N (-1)^k h_{N-k} \varphi(2t - k)$. For a time series measured at N equi-supposed time points, $h_0, h_1, \dots, h_n$ are constants called **filter coefficients**.

WAVEMETER: A device for measuring the geometrical spacing between successive surfaces of equal phase in an electromagnetic wave. For frequencies of up to about 100 MHz, a wavemeter consists of a tuned circuit with a suitable indicator to establish when resonance occurs. The tuned circuit usually includes a variable capacitor calibrated to read wavelengths and resonance is indicated by a current-detecting instrument. At higher frequencies a cavity-resonator in a waveguide is often used. The cavity resonator is fitted with a piston, the position of which determines the resonant frequency of the cavity.

WAX: 1. A class of hydrocarbons that are plastic (malleable) at normal ambient temperatures and include solid fatty substance of animal, vegetable or mineral origin. Wax melts above 45°C (318 K) to give a low viscosity liquid. It is insoluble in water but soluble in petroleum based solvent. Natural waxes such as beeswax and carnauba (a plant epicuticle wax) are solid substances made by some plants and animals for protection against water. Mineral waxes for candles and polishes are made from crude oil. 2. Also known as **earwax** or **cerumen**, it

is a yellowish waxy substance secreted in the ear canal of humans and other mammals. It protects the skin of the human ear canal, assists in cleaning and lubrication and also provides some protection from bacteria, fungi, insects and water. Excess or impacted cerumen can press against the eardrum and/or occlude the external auditory canal and impair hearing.

WAXING MOON: The increase in size of the Moon's illuminated surface that is visible from new Moon to full Moon.

WAXING CRESCENT MOON: Intermediate Moon phase that comes after new Moon and lasts until half of the Moon's surface is illuminated. At this phase, the Moon appears in the west shortly after sunset and has a curved shape similar to a banana or sickle.

WAXING GIBBOUS MOON: Moon that appears more than half illuminated but less than fully illuminated. At this phase, it rises between noon and sunset, setting in the early hours after midnight.

WCDMA (WIDEBAND CODE DIVISION MULTIPLE ACCESS): A third-generation (3G) standard that employs the code division multiple access (CDMA) technology over a wide frequency band to provide high-speed multimedia and efficient voice services. It was developed by Japan's NTT DoCoMo and formed the basis of its Freedom of Multimedia Access (FOMA) 3G Network. The WCDMA system is composed of 3 key parts: the user equipment (UE), UMTS terrestrial radio access network (UTRAN), and a core interconnecting network.

WEAK ACID: An acid such as acetic acid or carbonic acid that only partially ionises in aqueous solution. The pH of such acids lies between 3 and 6.

WEAK BASE: A chemical base that only partially ionises in aqueous solution. The pH of such bases, for example, ammonia lies between 8 and 10. As Bronsted-Lowry bases are proton acceptors, a weak base may also be defined as a chemical base in which protonation is incomplete resulting in a relatively low pH level compared to strong bases. Bases range from a pH of greater than 7 (7 is neutral, like pure water) to 14 (though some bases are greater than 14). The pH level has the formula: $\text{pH} = -\log_{10} [\text{H}^+]$. All organic bases are weak.

WEAK COLLISION: A collision between two molecules in which the amount of energy transferred from one to the other is not large compared to $k_B T$ (k_B is the Boltzmann constant and T the absolute temperature).

WEAK ELECTROLYTE: A substance that experiences little or no dissociation and conducts electricity only moderately. Weak acids and bases such as ethanoic acid and ammonia are weak electrolytes.

WEAK ERGODIC THEOREM (MEAN ERGODIC THEOREM): A weaker form of the Birkhoff ergodic theorem, in which one only obtains functions convergent in mean square.

WEAK FIELD LIGAND: Ligand that exerts a weak crystal or ligand field and forms high spin complexes with metals. π -donor ligands lead to a small Δ_o (the energy difference between 2 sets of d-orbitals called the splitting parameter, Δ_o) and are called weak- or low-field ligands. π -acceptor ligands lead to a large value of Δ_o and are called strong- or high-field ligands.

WEAK INEQUALITY: An inequality that permits the equality case. An example is example, $a \leq b$. If $a < b$ implies that $f(a) \geq f(b)$, f is a weakly decreasing function.

WEAK INTERACTION (WEAK FORCE; WEAK NUCLEAR FORCE): One of the four fundamental interactions of nature, along with strong interaction, electromagnetic force and gravitation that underlies some forms of radioactivity and certain interactions between subatomic particles. In the Standard Model of particle physics, it is due to the exchange of the heavy W and Z bosons. Its most familiar effect is radioactivity, specifically beta decay. It is a non-contact force, called weak because the typical field strength is 10–11 times the strength of the electromagnetic force and some 10–13 times that of the strong force, when forces are compared between particles interacting in more than one way. It is now known that the weak force has intrinsically the same strength as the electromagnetic force and the two are believed to be only different manifestations of a single electroweak force.

WEAK LAW OF LARGE NUMBERS: The sequence $\{a_n\}$ generated by calculating the average of the first n observations of a random variable x with finite variance, i.e., $a_n = \frac{1}{n} \sum_{i=1}^n x_i$ tends to the mean $\mu = E(x)$ with probability of 1 as $n \rightarrow \infty$, i.e., for any $\varepsilon > 0$ and any $\delta > 0$, there exists N for which $P\{|a_n - \mu| > \varepsilon\} < \delta$ for all $n > N$. These differ from the strong law of large numbers in which the convergence is pointwise.

WEAK NEUTRAL CURRENT: A very weak interaction that is independent of the electric charge of a particle. Particles can exchange energy through this mechanism, but other characteristics of the particles remain unchanged. This force is mediated by the Z-boson.

WEAK TOPOLOGY: The topology imposed on the underlying vector space such as a normed vector space with respect to its continuous dual by taking as a sub-base all open half spaces containing zero. This gives the weakest topology in which all the norm continuous linear functionals are continuous. A space is reflexive if and only if the unit ball is weakly compact or equivalently, by the Eberlein-Smulian theorem, if and only if it is weakly sequentially compact. 2. **Weak-star topology** is the corresponding topology placed on the dual normed space by requiring that a sequence or net $\{f_n\}$ converge to f , called weak-star convergence, if and only if, for every point x of the predual space, $f_n(x)$ tends to $f(x)$. In this topology the dual ball is weak-star compact.

WEAKLY COMPACT: Compact in the weak topology.

WEAKLY COMPACTLY GENERATED: A Banach space is compactly generated if it contains a weakly compact set, the closed linear span of which is the entire space. Every reflexive and separable Banach space is weakly compactly generated. A compactly generated Hausdorff space is a compactly generated space which is also Hausdorff.

WEAKLY CONVERGENT: A sequence or net $\{x_n\}$ is weakly convergent, if it is also convergent in the weak topology; $\{x_n\}$ converges weakly to x if and only if, for every continuous linear functional on the space, $f(x_n)$ tends to $f(x)$. Every strongly convergent sequence is also weakly convergent, but the reverse does not usually apply.

WEAKLY INTERACTING MASSIVE PARTICLE (WIMP): Hypothetical elementary particle that might make up most of the matter in the universe and that is also predicted to exist in supersymmetry theory, serving as one possible solution to the dark matter problem. The nature of the dark matter is not yet known, although many experiments are under way to try to discover it directly or indirectly. These particles interact through the weak force and gravity and possibly through other interactions no stronger than the weak force. Because they do not interact with electromagnetism they cannot be seen directly and because they do not interact with the strong nuclear force they do not react strongly with atomic nuclei.

WEAN: To introduce a mammal infant, either human or animal to food other than its mother's milk that will form its adult diet.

WEANLING: A male or female horse that is between the ages of six months to one year.

WEATHER: The state of the atmosphere at a particular place and time as regards heat, cloudiness, dryness, sunshine, wind, rain, etc. Weather is driven by air pressure, temperature and moisture differences between one place and another. These differences can occur due to the sun's angle at any particular spot, which varies with latitude. Its components are many which include sunshine, rain, cloud cover, winds, snow, flooding, blizzards, ice storms, thunderstorms, steady rains from a cold front or warm front, excessive heat, heat waves, etc.

WEATHER FRONT: A boundary between two air masses, one cold and dense, the other warm and less dense. Fronts are regions of uncertain weather, characterised by cloud and rain, variable humidity

and wind and low air pressure. A **warm front** is where warm air advances, displacing cold; the reverse happens in a **cold front**. An **occluded front** occurs often toward the end of a system of storms, when a mass of warm air is surrounded and forced upward by cold air.

WEATHER SATELLITES (METEOROLOGICAL SATELLITES): Spacecrafts that are used to monitor Earth's atmospheric conditions and provide data to help understand and predict global weather patterns. The first was TIROS-1 in 1960. Over the past four decades, the United States, Russia, Europe and other nations have launched weather satellites into both polar and geostationary orbits.

WEATHERED: 1. Seasoned or affected by exposure to the weather. 2. A weathered wood is one that is treated artificially or stained by the action of air, rain, etc. 3. A weathered rock is worn, disintegrated, or changed in colour or composition by the effects of weathering.

WEATHERING: The processes of physical, mechanical disintegration and chemical decomposition of rocks, minerals and immature soils at or near the Earth's surface. Physical weathering breaks up rock without altering its composition through the action of frost and extreme temperature. **Chemical weathering** decomposes rock by slowly altering its constituent minerals through dissolution of solid rock materials by water or carbonation, oxidation, hydration, etc. **Organic weathering** which may involve both mechanical and chemical is caused by plants and animals. For example, lichens can cause decomposition through the removal of nutrients and plant roots and burrowing animals may physically break up rocks.

WEAVING: A textile production method by which two distinct sets of threads or yarn are interlaced at right angles to form a fabric or cloth. Other methods of weaving include crocheting, knitting, braiding, felting and plaiting. The longitudinal threads are called the warp and the lateral threads are called the filling or weft. Many woven products are plain, satin or twill. They may be in one colour or simple pattern for the plain weave or artistic/decorative design for others.

WEB: 1. **World Wide Web; WWW** is a hypermedia computer-based information service system, distributed over a large number of computer sites that allows users to view web pages that may contain text, images, videos and other multimedia and navigate between them by using hyperlinks and retrieve information from documents containing links to the internet. It was developed in the early 1990s by CERN (Conseil Européen Pour la Recherche Nucléaire), an European Union organisation that carries out research into high energy particle physics, now known as European Laboratory for Particle Physics. 2. The skin joining the toes of some water birds such as ducks and geese and some water animals such as frogs or bats. 3. A network made of threads by a spider or other spinning creature.

WEB 2.0: A term that was introduced in 2004 and refers to the second generation of the World Wide Web. Web 2.0 introduces Internet technologies such as AJAX, Podcasts, RSS feeds, Weblogs, Wikis and other tools that help all users create their own content and to develop a more community driven website.

WEB ADDRESS (URL, UNIFORM RESOURCE LOCATOR): A standardised naming convention for addressing a specific Web site or file on the Internet. It does not have spaces or certain other characters and uses forward slashes to denote different directories. Examples are <http://www.bro.com/>, <http://web.ust.edu/> or <ftp://info.apple.com/>. Not all URLs begin with "http" and a portion of the URL is not required and many times can be left out.

WEB AUTHORING TOOL: In computing, software for creating Web sites in a desktop publishing format. The basic Web authoring tool is HTML, the source code that determines how a Web page is constructed and how it looks. Other programs, such as Java and VRML, can also be incorporated to enhance Web pages with animations and interactive features. Some other examples of Web Authoring Tools include NetObjects Fusion, Microsoft FrontPage and Macromedia Dreamweaver.

WEB BROWSER (BROWSER): An application program that provides a way to look at and interact with all the information on the World Wide Web. It uses HTTP (Hypertext Transfer Protocol) to make requests of

Web servers throughout the Internet on behalf of the browser user. Most browsers support e-mail and the File Transfer Protocol (FTP) but a Web browser is not required for those Internet protocols and more specialised client programs are more popular. Some common browsers are Microsoft Internet Explorer, Netscape Communicator and Apple Safari.

WEB CAMERA (WEBCAM): A video camera that streams images in real time to network (Internet) thus allowing everyone connected to the network to view either still pictures or motion video of a user or other object.

WEB FORUM: A website or section of a website that allows visitors to communicate with each other by posting messages. Most forums allow anonymous visitors to view forum postings, but require you to create an account in order to post messages in the forum. When posting in a forum, you can create new topics or threads or post replies within existing threads.

WEB HOST: A company in charge of hosting a web page and all its related content. The Web host stores all the pages of your website and makes them available to computers connected to the Internet.

WEB OF TRUST (WOT): An Internet browser service that helps identify websites that may be a potential harm to a computer by displaying a message over the website indicating it is not safe and should be avoided.

WEB PAGE (WEBPAGE): A document commonly written in HyperText Markup Language (HTML) that is accessible through the Internet or other network using a browser. A web page is accessed by entering a URL addresses and may contain text, graphics and hyperlinks to other web pages and files.

WEB PUBLISHING (ONLINE PUBLISHING): The process of creating Web site content on the Internet. It includes creating and uploading websites, updating webpages and posting blogs online. The published content may include text, images, videos and other types of media. In order to publish content on the web, the following are required: web development software, an Internet connection and a web server.

WEB SERVER: A computer system that hosts websites and used to deliver web pages and other content to multiple users. It runs Web server software, such as Apache or Microsoft IIS, which provides access to hosted webpages over the Internet. Most Web servers are connected to the Internet via a high-speed connection, offering OC-3 or faster data transmission rates. A fast Internet connection allows Web servers to support multiple connections at one time without slowing down. Any computer can be turned into a Web server by installing Web server software and connecting the machine to the Internet. The most popular Web server software in use worldwide is the Open Source Apache software: <http://www.apache.org>.

WEBBED: In botany, it describes an interlocking network of filaments, fibres, hairs or veins.

WEBCASTING: Broadcasting of audio, for example, from internet radio or video using the internet (World Wide Web).

WEBCRAWLER: An application that is used to systematically search other search engines, for the purpose of indexing.

WEBER: The SI unit of magnetic flux with the symbol Wb, which is equal to the flux that, linking a circuit of one turn, produces in it an emf of one volt as it is reduced to zero at a uniform rate in one second. It is equal to 1 Joule per ampere or 1 volt-second. The weber per square metre, called the tesla, with the symbol T, is the unit of magnetic flux density and a measure of the strength of a magnetic field in a given region. It is named after Wilhelm Weber (1804-91).

WEBER, WILHELM EDUARD (1804-91): German physicist, who became a professor at the University of Göttingen. In 1833, he and Karl Gauss worked on terrestrial magnetism and devised an electromagnetic telegraph. His main work was the development of a system of units that expressed electrical concepts in terms of mass, length and time. Since force was expressed in these dimensions, he

was then able to find his law of electric force. The principle was not very satisfactory because it did not conserve energy, but with it Weber publicised the view that matter was made up of charged particles held together by the force. This inspired the direction that physics took in the latter half of the century. Gauss had previously done this for magnetism. Both systems were adopted in 1881. The SI unit, magnetic flux is named after him. He introduced the absolute system of electrical units. The coulomb was once known as the weber; now the weber is a magnetic unit. In 1825, together with a brother, E. H. Weber, he wrote (1825) a book on wave motion.

WEBMAIL: A facility for creating, sending and receiving messages via the Internet. Webmail offers an alternative to using email software such as Outlook or Eudora. In order to use webmail you have to register with an Internet Service Provider (ISP) and you can then access their email service via your Web Browser.

WEBMASTER: A person in charge of maintaining a Web site. The jobs of a webmaster include writing HTML for Web pages organising the Web site's structure, responding to e-mails about the Web site and keeping the site up-to-date

WEBSITE (WEB SITE): An area on the World Wide Web where an organisation or individual stores a collection of pages of material, Web pages. The pages are usually interlinked with one another and with other websites. Every website has a unique Web Address or URL.

WEDDERBURN STRUCTURE THEOREM: The theorem that every semi-simple right Artinian ring is the direct sum of a finite number of simple right Artinian rings and a simple right Artinian ring is isomorphic to the ring of $n \times n$ matrices over some division ring, K for some positive integer, n .

WEDDERBURN'S THEOREM: In mathematics, it is a theorem that states that every finite domain is a field. It is named after Scottish mathematician, Joseph Henry Wedderburn (1882-1948).

WEDGE: 1. In mathematics, it is a triangular prism, a prism with the cross section of a triangle. 2. A polyhedral solid defined by two triangles and three trapezoid faces; it has five faces, nine edges and six vertices. 3. The symbol ($\wedge$), used to denote logical conjunction ("and"). 4. **Wedge sum** is a one-point union of a family of topological spaces. 5. In mechanics, it is an inclined plane, thick at one end and tapering to a thin edge at the other. It is usually made of metal, wood or any hard material and used for splitting, lifting or tightening, such as to secure a hammer head onto its handle. The wedge is considered one of the five simple machines. An example is an axe which is a wedge used as a cutting machine. 6. In meteorology, it is also known as ridge, it is an elongated zone of relatively high atmospheric pressure.

WEDGE PRODUCT (EXTERIOR PRODUCT): Of vectors, an algebraic construction used in Euclidean geometry to study areas, volumes and their higher-dimensional analogues. The exterior product of two vectors α and β is denoted by $\alpha \wedge \beta$. If α and β are differential k -forms of degrees p and q , respectively, then

$\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$. The magnitude of α and β can be interpreted as the area of the parallelogram with sides α and β , which in three dimensions can also be computed using the cross product of the two vectors. The **exterior algebra** or **Grassmann algebra** is the algebraic system whose product is the exterior product.

WEDGE PROJECTION: A stereochemical projection, roughly in the mean plane of the molecule, in which bonds are represented by open wedges, tapering off from the nearer atom to the farther atom. It is mainly used to illustrate the conformation of larger cycloalkanes, for example, cyclooctadecane.

WEED: Common name for unwanted and undesirable vegetation, growing where they are not wanted such as land under cultivation, where they compete with crops for water, light and nutrients. On pastures, they are disliked by most grazing animals as they can be poisonous. Some types are hosts of plant disease organisms or of insect and fungus pests. If found in their natural environments, weeds are sources of food for some insects and animals. They also enrich the earth, loosen hard-packed soils and help prevent erosion.

WEED CONTROL (WEED MANAGEMENT): The process of reducing weed growth to an acceptable level with minimum environmental pollution.

WEED KILLER (HERBICIDE): A chemical that kills or inhibits plant growth, especially of weeds or other undesirable vegetation. These chemicals come in different formulas and are almost as varied as the group of existing plants. Selective herbicides kill specific targets while leaving the desired crop relatively unharmed. Nonselective herbicides are not selective about which plants they kill. Any green plant that they contact will be injured or killed. Nonselective herbicides are classified as either **contact** or **systemic**. **Contact herbicides** such as sulfuric acid, diquat and paraquat; kill specific targets or part of the plant with which they come in contact with. **Systemic or translocated herbicides** such as amitrole, picloram and 2,4-D move within the plant from the point of application to other plant parts to destroy them. **Non-selective herbicides** can be used to clear waste ground, industrial sites, railways and railway embankments. Smaller quantities are used in forestry, pasture systems and management of areas set aside as wildlife habitat. Some plants produce natural herbicides in a process called **allelopathy**. An example of such a plant is the genus *Juglans* (walnuts).

WEEVIL: Common name for over 60,000 species of certain beetles in several families, mostly in the family Curculionidae. It is a small beetle with fore-wings, feeding on and infesting stores of grain, nuts and other seeds.

WEIERSTRASS APPROXIMATION THEOREM: The theorem that every continuous function defined on a closed interval $[a, b]$ can be uniformly approximated as closely as desired by a polynomial function.

WEIERSTRASS ELLIPTIC FUNCTIONS: The elliptic function

$$p(z) = \frac{1}{z^2} + \sum_{m,n} \left\{ \frac{1}{[z - \Omega(m, n)]^2} - \frac{1}{[\Omega(m, n)]^2} \right\} \text{ summed over non-zero}$$

integers, where $\Omega(m, n) = 2n\omega_1 + 2m\omega_2$, for independent periods $2\omega_1 + 2\omega_2$.

WEIERSTRASS FUNCTION: The function that is continuous everywhere, but differentiable nowhere because of its recursively constructed repeated oscillation $f(x) = \sum_{i=1}^{\infty} \lambda^{(s-2)^i} \sin(\lambda^i x)$, for s strictly

between 1 and 2 and $\lambda > 1$. The Hausdorff dimension of the graph has lower bound s ; although equality seems likely, this has never been proved. It is continuous but nowhere differentiable by reason of its recursively constructed repeated oscillation.

WEIERSTRASS M-TEST: The test for uniform convergence of a series based on the result that, for a family $\{f_n(x)\}$ of complex functions defined on a set E , $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on E as soon as one can find a summable real series, $\sum_{n=1}^{\infty} M_n < \infty$ such that $|f_n(x)| \leq M_n$ for all x in the set E .

WEIERSTRASS, KARL THEODOR WILHELM (1815-97): German mathematician, who was one of the founders of the modern theory of functions and known as the father of modern analysis. He made important contributions in many mathematical fields such as the theories of complex variables, power series, elliptic functions, quadratic forms and the calculus of variations. Many mathematical terms such as Weierstrass approximation theorem, Weierstrass-Casorati theorem, Weierstrass's elliptic functions, Weierstrass function and Weierstrass M-test have been named after him.

WEIGHT: 1. In mathematics, it is a value put on edges of a graph. 2. The process of attaching more importance to some observations than others. 3. In physics, it is the force acting on a body due to the gravitational field in which it is placed. The weight (W) of a body is determined by multiplying its mass (m) by the gravitational field strength (g): $W = mg$. The unit of weight is the newton (N).

WEIGHT TITRATION: A titration in which the amount of titrant required is determined in terms of the weight that must be added to reach the end point.

WEIGHT-DISTRIBUTION FUNCTION (MASS-DISTRIBUTION FUNCTION): A distribution function in which the relative amount of a portion of a substance with a specific value or a weight distribution range of values, of the random variable(s) is expressed in terms of mass fraction.

WEIGHTED GRAPH (EDGE-WEIGHTED GRAPH): A graph having a weight or number, associated with each edge. Some algorithms require all weights to be nonnegative, integral, positive, etc.

WEIGHTED MEAN (WEIGHTED AVERAGE): An average that can attach more importance (weight) to some observations than to others. It is an average calculated by taking into account not only the frequencies of the values of a random variable but also some other factors such as their standard deviations. The weighted average of observed data, where each of the values, v_i respectively occurs n_i times and each v_i has weighting of w_i is $\frac{\sum v_i n_i w_i}{\sum n_i w_i}$.

WEIGHTING: 1. In statistics, it is a factor by which some quantity is multiplied in order to make it comparable with others. 2. In taxonomy, it is assigning of greater or lesser importance to one character, as compared to another character, according to its known or assumed value. The initial choice of which characters to use in a classification is in itself a positive weighting process, termed selection weighting. Characters considered unreliable and consequently rejected, perhaps because they show too much environmental variation, are said to be given residual or rejection weighting.

WEIGHTLESSNESS (ZERO-G): Lack of gravity or zero gravity. If there is no gravitational force acting on a body, it is because gravitational force is cancelled out by equal and opposite acceleration or because the body is so far outside a planet's gravitational field that no force is exerted upon it. Weightlessness is the condition that exists for an object or person in free-fall when the effect of gravity is cancelled by inertia such as centrifugal force resulting from orbital flight. **Free-fall** is any motion that, in theory, is determined solely by gravitational forces. Examples are the Moon's motion around the Earth and an object dropping to the Earth's surface. The physical path of an inertial trajectory depends on the presence or absence of gravitation. If objects are far from a planet, star or other such massive body, so that they experience little gravitational interaction, they will experience weightlessness if they are not accelerating. Weightlessness in orbital motion is a sensation of weightlessness due to the spacecraft and all objects in it being in constant free-fall together as they circle Earth. Weightlessness in deep space is a sensation of weightlessness due to the minimal pull of gravity very far from any massive object.

WEIL'S DISEASE: A disease of humans and domestic animals caused by *Leptospira icterohaemorrhagiae* or *L. Canicola* commonly transmitted by urine of infected cattle or rats. The disease can range from very mild and symptomless to a more serious, even life threatening form, that may be associated with kidney (renal) failure.

WEINBERG-SALAM MODEL (WS MODEL): The unification of the electromagnetic and weak interactions described by the Weinberg-Salam theory, for which Abdus Salam, Sheldon Glashow and Steven Weinberg were awarded the Nobel Prize in Physics in 1979.

WEISMANNISM (WEISMANN BARRIER): The theory that all heritable characteristics arise in the germ plasm and that acquired characteristics cannot be inherited. The content of the reproductive cells are passed on unchanged from one generation to the other unaffected by any changes undergone by the rest of the body. The theory is very important as it has implications for human gene therapy.

WEISSENBERG TECHNIQUE: A technique used to overcome the problem of overlapping reflections in the identification of the symmetry and the dimensions of a unit cell in x-ray crystallography. In this technique, a screen is placed in front of the film, allowing only one

set of reflections to be exposed. The Weissenberg technique produces distorted photographs, but this can be overcome by having a coupling between the motions of the crystal and the film.

WEISZ RING OVEN: A device for vaporisation of solvent from filter paper, leaving the solute in a ring (circular) shape. It is used for qualitative analysis of very small samples.

WEIGHTING: 1. In statistics, it is a factor by which some quantity is multiplied in order to make it comparable with others. 2. In taxonomy, it is assigning of greater or lesser importance to one character, as compared to another character, according to its known or assumed value. The initial choice of which characters to use in a classification is in itself a positive weighting process, termed selection weighting. Characters considered unreliable and consequently rejected, perhaps because they show too much environmental variation, are said to be given residual or rejection weighting.

WELDING: A fabrication or sculptural process that joins materials, usually metals or thermoplastics, by causing coalescence. It is often done by applying heat, sometimes with pressure or with an intermediate or filler metal having a high melting point to form a pool of molten material that cools to become a strong joint. The various energy sources used for heating include: flame, electric arc, a laser, an electron beam, friction and ultrasound. The most common gas welding process is oxyfuel welding, also known as oxyacetylene welding. Welding is a hazardous process and requires a trained person, who should take precautions to avoid fire, burns, electric shock, vision damage, inhalation of poisonous gases and fumes and exposure to intense ultraviolet radiation. Soldering and brazing are not the same as welding. They involve melting a lower-melting-point material between materials to form a bond between them, without melting the materials. A person trained to weld is called a **welder**.

WELFARE CODE: An official set of rules for making sure that animals are healthy and happy in a particular situation, for example, in quarantine or when being transported.

WELL: An artificial excavation made to extract water, oil, gas, brine or other fluid substance from the earth. An example is artesian well which is drilled through impermeable rocks into strata where water is under enough pressure to force it to the surface without pumping.

WELL-CONDITIONED: 1. In mathematics, a well-conditioned problem is a problem that has a low condition number. 2. Of a computation, being numerically stable.

WELL-DRAINED SOIL: 1. Soil in which water infiltrates at a reasonably higher infiltration rate (measured by the depth, in mm of the water layer that the soil can absorb in an hour). Such soils include deep, loamy soil and sloping sites. Soils that have high clay content, depressions, or sites with high water tables, underlying rock not well drained.

WELL-FORMED: In logic, often abbreviated wff, a well-formed formula or expression is one that is grammatically correct and constructed in accordance with the formation rules of a particular formal system.

WELL-ORDERED SET: A totally ordered set having the property that every non-empty subset of its field has a least member under the relation. For example, less than is well-ordered on the natural numbers, but not on the reals, since an open set has no least member.

WELL-ORDERING PRINCIPLE (ZERMELO'S THEOREM): 1. In logic, also called well-ordering theorem, it is the non-constructive set theoretic result, equivalent to the axiom of choice or Zorn's lemma, that for any set there is a binary relation under which it is well-ordered. 2. The principle that the positive integers are well-ordered.

WELL-POSED PROBLEM: A problem that is well formulated and can be shown to satisfy the following conditions:

- (i) that the solution exists;
- (ii) that the solution is unique; (iii) that the solution varies continuously with perturbation of the data. If the problem does not meet all of the conditions, it is **ill-posed**, although it may still be soluble.

WELSH: Refers to a breed of pigs, white in colour, with lop ears. It is one of the older breeds of British pigs.

WELSH BLACK: A hardy dual-purpose breed of cattle, the product of the northern Anglesey strain bred with the Castle-martin strain. Welsh blacks produce a reasonable milk yield and very lean meat.

WELSH MOUNTAIN: A hardy breed of sheep, well adapted to wet conditions. They are small with white faces, very fine fleece, with only the rams having horns.

WESTERGREN SEDIMENTATION RATE (ERYTHROCYTE SEDIMENTATION RATE; SED RATE; SEDIMENTATION RATE):

In medicine, it is a non-specific blood test that is a measure of the rate at which red blood cells (erythrocytes) settle out of suspension in blood plasma. ESR is useful in detecting inflammation associated with conditions such as serious infections, cancers and autoimmune diseases.

WESTERLIES: Prevailing winds from the west that occur in both Northern and Southern Hemispheres between latitudes 35° and 60°. The westerlies of the Northern Hemisphere blow from the south-west; those of the Southern Hemisphere blow from the north-west. They are variable and produce stormy weather.

WESTERN BLOTTING (PROTEIN BLOTTING; WESTERN BLOT):

A method of detecting and measuring very small amounts of a particular protein in tissue samples or cells based on their ability to bind to specific antibodies. The sample is subjected to electrophoresis on SDS-polyacrylamide gel to separate constituent protein. The resultant protein bands are then transferred (blotted) to a nitrocellulose sheet and then treated with radioactive antibodies or antibodies coupled to a fluorescent dye or an easily detectable enzyme that are specific to the protein of interest.

WESTON CELL (CADMIUM CELL): A type of primary voltaic cell, invented by Edward Weston in 1893, that produces a highly stable voltage of 1.0186 volts at 20 °C (293 K), which is used as a laboratory standard for calibration of voltmeters. It is usually made in an H-shaped glass vessel with a mercury anode covered with a paste of cadmium sulfate and mercury(I) sulfate in one leg and a cadmium amalgam cathode covered with cadmium sulfate in the other leg. The electrolyte, which connects the two electrodes is a saturated solution of cadmium sulfate. In certain cells, sulfuric acid is added to stop the hydrolysis of mercury sulfate. It was adopted as the International Standard for EMF between 1911 and 1990.

WET ASHING: The conversion of an organic compound into ash by treating the compound with nitric or sulfuric acid.

WET BULB TEMPERATURE: The wet-bulb temperature is the lowest temperature that can be reached under current ambient conditions by the evaporation of water only; it is the temperature felt when the skin is wet and exposed to moving air. Wet-bulb temperature is largely determined by both actual air temperature (dry-bulb temperature) and humidity, the amount of moisture in the air.

WET-BULB DEPRESSION: The difference between wet-bulb and dry-bulb temperatures that is used to find relative humidity and dew point temperatures. It is measured by a psychrometer. If there is 100% humidity, dry-bulb and wet-bulb temperatures are identical, making the wet-bulb depression equal to zero in such conditions.

WET CELL: A battery that has a liquid electrolyte (such as sulfuric acid). Wet cells may be primary cells (non-rechargeable) or secondary cells (rechargeable). The rechargeable type is an electrochemical cell in which the chemical reaction producing the e.m.f. (electromotive force) is reversible and the cell can therefore be recharged by passing a current through it. It has two terminals for supplying power to an external circuit. The one marked "+" is the cathode (positive) and that marked "-" is the anode or negative. The negative terminal is the source of electrons that supply current to a load. When a load is connected to the battery a chemical reaction takes place in the electrolyte moving ions within to supply current to the terminals and to an external circuit. An example of a rechargeable battery is lead-acid battery used in the ignition system of most motor vehicles.

WET ROT: A plant disease in which there is disintegration of tissues and release of cell fluids. Brown rot of stored fruits, such as plums and apples, is caused by fungi of the genus *Sclerotinia*. Wet rot of structural timber is caused by various fungi including *Coniophora puteana* and *Poria vaillantii*.

WET-AND-DRY BULB HYGROMETER: An instrument consisting of two thermometers, used in the measurement of the moisture content of air or other gases. The bulb of one of the thermometers is kept wet by a thin piece of clean muslin cloth wetted uniformly with distilled water or is otherwise coated with a film of distilled water and shows a lower temperature. The relative humidity is obtained from the two temperatures using tables.

WETHER: Castrated male sheep at an early age before secondary sex characters have developed.

WETLAND: A lowland area, such as a marsh, swamp or peatland, where the soil near the surface is saturated or covered with water, especially when regarded as the natural habitat of wildlife. Wetlands typically occur in low-lying areas that receive fresh water at the edges of lakes, ponds, streams and rivers or salt water from tides in coastal areas protected from waves. The different plant species of a wetland provide habitat for varied animal communities.

WETTABILITY: The ability of any solid surface to be wetted when in contact with a liquid, i.e. the surface tension of the liquid is reduced so that the liquid spreads over the surface.

WETTED: Pertaining to a material that has accepted water or other liquid and withers on its surface or within its pore structure. It is the ability of a liquid to maintain contact with a solid surface, resulting from intermolecular interactions when the two are brought together. The degree of wetting (wettability) is determined by a force balance between adhesive and cohesive forces. Wetting is important in the bonding or adherence of two materials. Wetting and the surface forces that control wetting are also responsible for other related effects, including so-called capillary effects.

WETTING AGENT: A chemical, such as detergent or soap that lowers the surface tension of water. Wetting agents are added to fungicide, herbicide and insecticide sprays to increase the area of the spray droplets in contact with the plant surface or insect body.

WHALE: Any of various marine mammals of the order Cetacean that breathe through a blowhole on the top of their heads and have front flippers, no hind limbs and flat horizontal tails. Unlike fish, which move their vertical tail fins from side to side when they swim, whales move their horizontal tail fins up and down to propel themselves through water. They maintain a warm and constant body temperature of about 37°C and breathe air with lungs. The blue whale (*Balaenoptera musculus*) is one of the largest animals that has ever lived and can reach a length of over 24 metres and a weight of 150 metric tons. Their bodies are insulated by thick layers of fatty blubber beneath their skins and many species live in social groups, communicating by sound. Some are hunted for their oil and flesh.

WHALE SHARK: A slow-moving filter-feeding fish (*Rhincodon typus*) which is the largest fish in the world at a length of 12.7 metres and weighing 21.5 metric tons. Its head is broad and flattened with a large terminal mouth, miniscule teeth and large gill slits. The eyes are small and located just behind the mouth on each side of the head. There are three prominent ridges along the back. The first dorsal fin is larger than the second dorsal fin and is set midway back on the shark. It feeds chiefly on plankton strained by its gill rakers.

WHALEBONE (BALEEN): The elastic, horny material forming the fringed plates hanging down from the upper jaw on each side of the mouth of baleen or toothless whales, forming a sieve for straining plankton from the water. When water containing plankton enters the open mouth and is expelled with the mouth slightly closed, the plankton is trapped on the baleen plates.

WHAT YOU SEE IS WHAT YOU GET (WYSIWYG): In computer science, it is an application that enables one to see on the display screen exactly what will appear when the document is printed.

WHEAT (BREAD WHEAT; COMMON WHEAT; SPRING WHEAT; FIELD WHEAT; *Triticum aestivum*): A monocotyledonous plant which belongs to the grass family, Graminae (Poaceae) and the third most produced cereal after maize (*Zea mays*) and rice (*Oryza sativa*). It is a cultigen and known only in cultivation in temperate climates with about 5,000 cultivars currently in use. The monoecious plant, which is 1 metre tall, develops several side shoots called tillers from a crown just below ground level. Each tiller bears axillary leafy culms (stems) that comprise five to seven nodes with three to four alternate flat leaves. It produces a spike on each of its culms. The spikes are the flowering and fruiting part. The uppermost or flag leaf, subtends an inflorescence consisting of perfect flowers, which are self-pollinated. Each spikelet is a condensed reproductive shoot consisting of two subtending sterile bracts or glumes. The glumes enclose two to five florets that are borne on a short axis or rachilla. The florets contain three stamens with large anthers and the pistil, which comprises a single ovary, with a single ovule, two styles and two branching plumose stigmas at the end of each style. The wheat grains are oval and some almost spherical to long, narrow and flattened shapes. The grains are red in colour, although some varieties have white, purple, black, brown or green/grey colours. Wheat flour is used for making bread, cake, pastries, crackers, biscuits and muffins. The straw is used for fuel, livestock bedding and fodder, thatching and basket-making.

WHEATSTONE BRIDGE: An electrical circuit invented by Samuel Hunter Christie in 1833 and improved and popularised by Sir Charles Wheatstone in 1843 for measuring the value of an unknown resistance by balancing two legs of a bridge circuit, one leg of which includes the unknown component. It consists of an electrical circuit, three known resistances and a galvanometer. If R_x is the unknown resistance to be measured; R_1 , R_2 and R_3 are resistors of known resistance and the resistance of R_2 is adjustable. If the ratio of the two

resistances in the known leg $\left(\frac{R_2}{R_1}\right)$ is equal

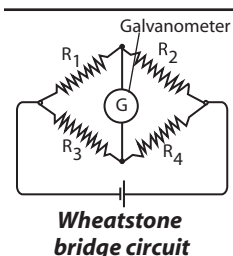
to the ratio of the two in the unknown leg $\left(\frac{R_x}{R_3}\right)$, then the voltage between the two

midpoints (B and D) will be zero and no current will flow through the galvanometer V_g . R_2 is varied until this condition is reached. The direction of the current indicates whether R_2 is too high or too low. Detecting zero current can be done to extremely high accuracy. Therefore, if R_1 , R_2 and R_3 are known to high precision, then R_x can be measured to high precision. Very small changes in R_x disrupt the balance and are readily detected. At the point of balance, the ratio of $\left(\frac{R_2}{R_1}\right) = \left(\frac{R_x}{R_3}\right)$.

$R_x = \left(\frac{R_2}{R_1}\right) R_3$. The various types of the Wheatstone bridge include the

metre bridge. In this case, a wire 1 metre in length of uniform resistance is joined to the top of a board alongside a metre rule. A sliding contact is run and the wire is analogous to R_2 and R_3 , until the galvanometer registers zero. Most types, however use one or more rotary rheostats to provide the needed variation.

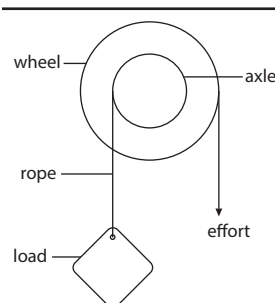
WHEATSTONE, SIR CHARLES (1802-75): British physicist and inventor, who was a musical instrument-maker in London. He studied acoustics and optics, inventing a stereoscope in 1838. His



most important work, done with William Cooke (1806-79), was the development of an electric telegraph, which they achieved in 1837. He popularised a method for the measurement of electrical resistance using a method now known as the Wheatstone bridge, to which he gave his name, although he did not invent it. He invented many other devices, including an automatic transmitter, an electric recording apparatus and an automatic telegraph. He is also credited with the invention of the concertina and with improving the stereoscope and the dynamo.

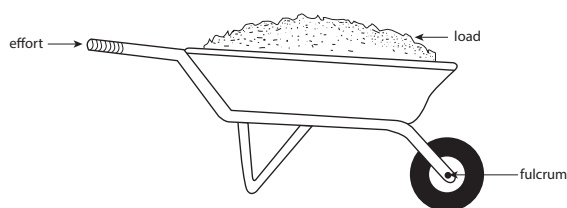
WHEEL: A circular framework or disc capable of turning on an axle. It is used for various other purposes in machines such as carts, cars and bicycles to provide motion on a track or road.

WHEEL AND AXLE: A simple machine with a rope wound round an axle connected to a larger wheel with another rope attached to its rim so that they rotate as a unit on a common axis. The principle of operation is the same as that of the lever, in that for static equilibrium, the summation of torques about the axis of rotation equals zero. It usually consists of a cylindrical drum and a wheel mounted on the same axle with ropes wound around each. Pulling on the wheel rope (applying an effort) lifts a load attached to the axle rope. The larger the wheel, the greater the force over the distance of the axle. It is often used to raise or lower loads. Where flexible members, such as ropes, have been firmly attached to a wheel and drum and the machine is mounted on frictionless bearings, $F_1R_1 - F_2R_2 = 0$.



Wheel and axle

WHEELBARROW: A small vehicle with one wheel and two handles at the rear for moving small loads. It is designed to distribute the load between the wheel and the operator, so that loads can be conveyed much easier than being carried by the operator alone. It is classified as a second class lever. It is commonly used in construction and gardening.



Wheelbarrow

WHIPCORD WESTERN RED CEDAR (WESTERN RED CEDAR; Thuja plicata 'Whipcord'): An evergreen non-flowering tree of family Cupressaceae, found in North America. It grows in lush forest and mountainsides. It is sometimes referred to as riparian tree since it can grow in forested swamps and stream banks. It is a shade loving plant capable of reproducing under dense shade. The leaves have opposite arrangement and are largely aromatic, with scent reminiscent of pineapple when crushed. Because of its coniferous nature it produces cones that are slender. It is used for timber as well as ornamentals for screens and hedges in gardens and packs. The bark is easily removed for making mats, rope and cordage, rain hat, clothing, etc.

WHIPLASH FLAGELLUM (ACRONEMATIC FLAGELLUM): A threadlike projection arising from motile algal and fungal cells that has a smooth surface. Internally the fibrillar structure is typical of the flagella and cilia of motile eukaryotic cells.

WHIRLWIND: 1. A rapidly rotating column of air, often synonymous with a tornado. On a smaller scale it produces the dust-devils seen in deserts. 2. The first electronic digital computer used in a real-time application and the first to use magnetic core memory.

WHISKER: 1. A strong very thin fine filamentary crystal with extraordinary shear strength and unusual electrical or surface properties. 2. Also known as **vibrissa**, the long stiff hair growing near the mouth of some mammals such as mice, cats and rabbits. In animals, whiskers or vibrissae stimulate nerve centres in the skin when they vibrate and are used as organs of touch.

WHISTLING PINE: A common, tall pine/conifer-like tropical tree that grows up to 20.30 metres in height. It has soft, wispy appearance, with hanging green branchlets and cone-like fruits. The leaves are reduced to rings of scales around long, slender and grooved branches that are somewhat like pine needles. This helps to reduce transpiration and conserve water. It is a nitrogen fixing tree, used for erosion control by virtue of the fact that it serves as a wind-break. Since it is very strong and needs little processing, it is ideal for posts, rough house construction, fencing, piling and roofing shingles

WHISTLER: 1. An electromagnetic wave of audio frequency produced by atmospheric disturbances such as lightning, having a characteristically decreasing frequency responsible for a whistling sound of descending pitch in detection equipment. The lightning produces long-wavelength electromagnetic waves, which follow the magnetic lines of force on the earth from the point where the lightning strikes to the antipodal point on the earth and back again several times. 2. A horse having a respiratory disease characterised by wheezing.

WHITE: The colour of fresh snow or common salt. It is the achromatic colour of maximum lightness. It is the colour of objects that reflect nearly all light of all visible wavelengths. The perception of white is a response to maximum stimulation of the retina that appears always to depend on contrast.

WHITE ALKALI SOIL (SOLONCHAK): A type of intrazonal infertile soil with a high salt content (halomorphic). It is found in arid and semiarid inland regions and in areas where sea spray has blown inland resulting in the deposition of salt.

WHITE ALLOY: Any of various alloys that are used as a base for plated silverware ornaments or novelties, as well as any of several lead-base or tin-base alloys used for bearings, jewellery, miniature figures, fusible alloys, etc. Some of the metals that make up white metal alloy are antimony, tin, lead, cadmium, bismuth and zinc.

WHITE ANT (TERMITE): A light-coloured social insect of the order Isoptera that forms large colonies. They have chewing mouthparts and live mostly in warm or tropical regions and feed on wood, from which they obtain cellulose. They are highly destructive to trees and wooden structures. Some species build huge mounds of up to 12 metres tall to house their colonies.

WHITE BALANCE (WB): A feature of many digital cameras and video cameras, which is used to accurately balance colour. It defines what the colour white looks like in specific lighting conditions, which also affects the hue of all other colours. When white balance is off, digital photos and recordings may appear to have a certain hue cast over the image. For example, fluorescent lights may cause images to have a greenish hue, while pictures taken on a cloudy day may have a blue tint.

WHITE BLOOD CELL (LEUCOCYTES; WHITE BLOOD CORPUSCLE): Colourless cells with a nucleus and cytoplasm, found in blood and lymph. They are capable of separating into a thin white layer when whole blood is centrifuged. They have independent

amoeboid movement and play a role in the body's defences as well as give immunity against diseases. There are two types: those without granules in the cytoplasm and those with granular cytoplasm and include neutrophils, eosinophils, basophils, lymphocytes and monocytes.

WHITE DWARF: A small star, which is the remnant of one that has used all its fuel and shrunk under its gravitational force, yet continues to radiate light from thermal energy trapped in it during its collapse. This is the final luminous phase in the evolution of low- to medium-mass stars. Its size is approximately that of Earth but with the mass of the Sun.

WHITE HOLE: A region of space-time that is the time reversal of a black hole; or a hypothetical hole in outer space from which energy, stars and other celestial matter emerge or explode.

WHITE MATTER: Part of the tissue composed mainly of the fibres or axons of nerve cells, most of which are covered by whitish, fatty myelin sheaths that form the central nervous system of vertebrates. The axons in the white matter are responsible for conducting information from and to the nerve cell bodies that are concentrated in the grey matter.

WHITE MUSCLE: A rapid or fast-twitch muscle, in which large pale fibres predominate. In a white muscle, mitochondria and myoglobin are relatively sparse.

WHITE MUSCLE DISEASE (NUTRITIONAL MYOPATHY; NUTRITIONAL MUSCULAR DYSTROPHY): Muscle degeneration in lambs and calves, caused by lack of selenium and/or vitamin E.

WHITE NOISE: Random signal in which the energy per unit bandwidth is constant for the whole frequency range of interest or the signal contains equal power within a fixed bandwidth at any centre frequency.

WHITE ORCHID (*Bauhinia aculeata*): A deciduous to semi-evergreen tropical plant in the family Leguminosae, which is about 9 metres tall. The ascending branches are twisted and droop at the ends. The leaves are large, bilobed and dark green. The white orchid tree becomes uniform and symmetrical as it grows older with dark brown seed pods which are flat. The white orchid trees are planted for shade, reclamation of land, parking lots, tree lawn and highway median.

WHITE ROT FUNGUS: Fungus that attacks lignin, along with cellulose and hemicellulose, leading to bleaching of the infected wood.

WHITE SPIRIT: A colourless, transparent liquid derived from petrol. It is a mixture of aliphatic and alicyclic C7 to C12 hydrocarbons with a maximum content of 25% of C7 to C12 aromatic hydrocarbons. It is poisonous if ingested. It is used as an extraction solvent, as a cleaning solvent, as a degreasing solvent and as a solvent in aerosols, paints, wood preservatives, lacquers, varnishes and asphalt products.

WHITELIST: A list of entities approved for authorised access or privileged membership to enter a specific area in the computing world. When a whitelist is used, all entities are denied access, except those included in the whitelist. The opposite of a whitelist is a blacklist, which allows access from all items, except those included the list.

WHITEWASH: A suspension of calcium hydroxide in water, used to whiten walls, fences or other structures.

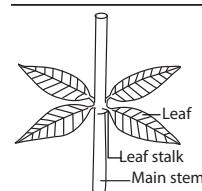
WHOIS: In computer science, it is an Internet service that searches database of registered domain name or IP address. It can also tell you information such as whether a domain name is available or has already been taken. A special application, also called whois is needed to search the database using Allwhois.com or BetterWhois.com.

WHOLE NUMBER: Counting numbers starting from zero (e.g., 0, 1, 2, 3, 4, 5, 10, 200, etc.) that does not contain a decimal or fraction.

WHOOPING COUGH (PERTUSSIS): A highly contagious bacterial disease of the respiratory system caused by the bacterium *Bordetella*

pertussis that causes violent coughing spasms followed by sharp, shrill inhalations. It mainly affects children.

WHORL: 1. One of the three main groups of human fingerprint patterns or a series of concentric circular or elliptical ridges in the pattern of lines on the gripping surface of a finger or thumb. 2. In botany, it is a circular arrangement of three or more leave petals radiating from a single node, as found in *Galium aparine* (common bedstraw). 3. A turn or coil in a mollusc's shell. 4. Anything in the shape of a curl, spiral or coil.



Whorled leaves

WI-FI PROTECTED ACCESS (WPA): An improved data encryption standard introduced with the 802.11i wireless standard in 1999 that helps reduce the security vulnerabilities with WEP by using either Temporal Key Integrity Protocol (TKIP) and Advanced Encryption Standard (AES) standard technologies.

WICKER-WORK-TRAP: An instrument used for harvesting fish. It is usually made of palm fronds, and used often along water bodies such as rivers, streams and lakes.

WIDE AREA INFORMATION SERVER (WAIS): Software tool formerly used in early internet search engines for indexing large amounts of information and search for and retrieve it from a range of databases on remote computers including the Internet. It has now been replaced by programs such as Google.

WIDE AREA NETWORK (WAN): A network of computers such as the Internet located at geographically separate sites. Many wide area networks span long distances via telephone lines, fibre-optic cables or satellite links. They can also be composed of smaller LANs that are interconnected. For home routers the port the router connects to the Internet connection is often labelled as a WAN, Network or Internet port, since it is what allows home network to communicate with the Internet network.

WIDGET (GADGET): An additional feature or extension to a software program, operating system GUI or web page, which enables additional features; improving the user's overall experience. An example is the Google advertisement, which is produced by placing a small piece of code on the web page.

WIEDEMANN-FRANZ LAW: An empirical law of physics, which states that the ratio of the thermal conductivity (K) of any pure metal to its electrical conductivity (σ) is approximately constant at a given temperature (T). It is expressed as $K/\sigma = L \times T$.

WIEN FORMULA (WIEN'S RADIATION LAW; WIEN'S DISPLACEMENT LAW): A formula for the spectral distribution of radiation from a blackbody. It states that the wavelength distribution of thermal radiation from a black body at any temperature has essentially the same shape as the distribution at any other temperature, except that each wavelength is displaced on the graph. Apart from an overall T^3 multiplicative factor, the average thermal energy in each mode with frequency ν only depends on the ratio ν/T . Restated in terms of the wavelength, $\lambda = c/\nu$, the distributions at corresponding wavelengths are related, where corresponding wavelengths are at locations proportional to $1/T$. From this general law, it follows that there is an inverse relationship between the wavelength of the peak of the emission of a black body and its temperature when expressed as a function of wavelength (frequency gives a different peak as noted later): $\lambda_{\max} = b/T$, where $\lambda_{\max}$ is the peak wavelength, T is the absolute temperature of the blackbody and b is a constant of proportionality called **Wien's displacement constant**, which is equal to $2.8977685(51) \times 10^{-3} \text{ m}\cdot\text{K}$.

WIENER PROCESS: In probability, it is a stochastic process that models Brownian motion; a family of real-valued random variables X_t ($t \geq 0$) with $X_0 = 0$ almost everywhere, such that each $X_{t+s} - X_t$ ($s, t \geq 0$) is normally distributed with mean zero and variance

s , while for $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $X_{n+1} - X_n$ are independent for $0 \leq i < n$. It is named after the English number theorist, Norbert Wiener (1894-1964).

WIFI (WIRELESS FIDELITY): A term used to refer to a networking technology that allows computers and other devices to communicate without a Local Area Network (LAN) that uses cable connections to connect to the internet.

WIGNER ENERGY (WIGNER EFFECT; DISCOMPOSITION EFFECT): The displacement of atoms in a solid caused by neutron radiation. The effect is of most concern in neutron moderators, such as graphite and boron that are used to slow down neutrons. The material surrounding the moderator receives a much smaller amount of neutron radiation and from slower neutrons and is not as worrisome. The Wigner effect is possible because neutrons usually have energies in excess of the threshold energy of 25eV required to displace atoms in a crystal lattice. A high energy neutron will create a displacement cascade in a crystal by way of elastic collisions. Some of the displaced atoms come to rest in a non-ideal location and are called Wigner Energy. Accumulated interstitials may release all their energy suddenly, creating a temperature spike, a situation which poses tremendous risk in the operation of low temperature nuclear reactors.

WIGNER NUCLIDES: The most important class of mirror nuclides, comprising pairs of isobars with odd nucleon numbers in which the atomic number and the neutron number differ by one. Examples are ^3H and ^3He .

WIGNER RULE (SPIN CONSERVATION RULE): For both radiative and radiationless transitions, the principle that transitions between terms of the same multiplicity are spin-allowed, while transitions between terms of different multiplicity are spin forbidden.

WIGNER-SEITZ CELLS: A polyhedron in a crystal that is bounded by planes formed by perpendicular bisectors of bonds between lattice sites. The concept has been used extensively in the theory of solids.

WIGNER-WITMER RULES (CORRELATION RULES): A set of results found by applying group theory that states which molecular electronic states can exist, starting from the electronic states of the isolated atoms. The rules were stated for diatomic molecules by Hungarian-born US physicist Eugene Wigner (1902-95) and E. E. Witmer in 1928 and were subsequently extended to polyatomic molecules. These rules are also known as correlation rules since they involve the correlation between atomic electronic states and molecular electronic states. They are useful in analysing the electronic spectra of molecules.

WIGNER'S RULE: A result concerning the quantum statistics of systems with more than one fermion. If there are N fermions in the system and N is odd, then the total system is a **fermion**; but if N is even, the total system is a **boson**.

WIKI: The term refers to a Web site that allows users to add and update content on the site using their own Web browser. This is made possible by Wiki software that runs on the Web server. Wikis end up being created mainly by a collaborative effort of the site visitors. An example of a large Wiki is the Wikipedia, a free encyclopedia in many languages that anyone can edit.

WILCOXON TEST (WILCOXON SIGNED RANKS TEST): In statistics, it is a one-sample test upon the ranks of some given data, used in examining whether or not the population from which the data are drawn has a given median. For example, a test for the relative level of the scores of the same subjects under two experimental conditions.

WILD FLOODING IRRIGATION: A method of irrigation in which water is released at high points in the field and distribution is uncontrolled.

WILD POINSETTIA (MEXICAN FIREPLANT; MILK WEED; *Euphorbia heterophylla*): A hardy, herbaceous, erect annual plant belonging to the Euphorbiaceae family that grows to about 20 to 80 centimetres tall, producing milky latex in all parts. It is native to Central

and South America, but is now distributed in most of the tropical and subtropical regions of the world; occurring in cultivated areas, pastures and wastelands. The stem is simple or branched with hollow stems and angular ribs. The leaves are usually arranged oppositely on the lower parts of the stem and just below the flower clusters and alternately arranged along most of the stem. The leaves directly beneath the flowers are usually entirely green in colour, but their bases may be a slightly different colour. The inflorescence consists of densely-clustered cyathia in axillary or terminal cymes. The cyathia are cup-like involucre, each containing several tiny male flowers, which are reduced to stamens; and one female flower consisting of a very large stalked ovary topped with a stigma. Each of the involucre has a smaller rounded cup-like projection, which is the floral nectary or gland. The plant bears three-lobed glabrous capsules, 3-4 millimetres long and 5-6 millimetres across, each containing three seeds. The plants grow very quickly and are very competitive with crops and are especially troublesome to soybeans, cowpeas, maize and sugarcane. In Ghana, it occurs as a weed of cassava; and in Nigeria, as that of cowpeas and soybeans.

WILD SAGE (RED SAGE; WHITE SAGE; TICKBERRY; *Lantana camara*): A small perennial shrub in the family Verbenaceae that can grow up to about 2 metres in height and forms dense thickets in different environments. It is native to Central/South America and West Africa and can be found in some tropical and sub-tropical countries worldwide including Spain, Portugal, the Middle East, India, Australia, New Zealand and parts of Africa. The leaves are opposite, ovate, simple and arranged oppositely on the stem. They have a strong odour when crushed. The small tubular shaped red, yellow, white, pink or orange flowers have four petals and are borne in dense clusters on the axils near the top of the stem. It produces small fleshy berry-like fruits which tend deep purple. Each fruit contains one seed. Reproduction is by both vegetative and sexual and it is considered a weed. It is an invasive species and a good competitor to other desirable species and can cause problems if not checked in agriculture areas because it forms dense thicket which could reduce the productivity of farm lands. It is a poisonous plant and both the unripe fruits and the foliage are very toxic and can cause damage to livestock. However, the leaves can be boiled to treat swellings and pain of the body. Its bark is astringent and used as a lotion in cutaneous eruptions and leprosy ulcers. In some areas, it is grown as a hedge plant. Its stalks are used as for making paper.

WILD TYPE: The naturally occurring gene for a particular character that is typical of most individuals of a given species, as distinct from new genes which arise by mutation or the form of an organism, strain or gene that results from natural breeding, as distinct from mutant forms or those resulting from selective breeding. Wild type alleles are often dominant. It is possessed by most members of a population in their natural environment.

WILDCARD: A term used in computing to refer to a character that can be substituted for zero or more characters in a string. Wildcards are commonly used in computer programming, database SQL search queries and when navigating through DOS or Unix directories via the command prompt.

WILDERNESS: An area of uncultivated and uninhabited land, which is usually, located some distance from towns and cities. It is an uninhabited region left in its natural state.

WILDLIFE: Non-domesticated plants, animals and other organisms living in a natural undomesticated state. Wildlife of distinct forms can be found in all ecosystems such as deserts, forests, rain forests, plains, grasslands, including the most developed urban sites. Some wildlife species are in danger of extinction. The primary causes of species extinction or endangerment are habitat destruction, over exploitation (such as plant collecting, hunting and trade in animal parts), damage caused by exotic plants and animals introduced into

an area and pollution. Of these causes, direct habitat destruction threatens the greatest number of species.

WILDLIFE CONSERVATION: Preservation and management of wildlife populations as a recreational and ecological resource. In an effort to protect global biodiversity and encourage the study, restoration and sound management of endangered species, international organisations such as the International Union for Conservation of Nature (IUCN) and the World Conservation Monitoring Centre (WCMC) maintain a global list of endangered and vulnerable animal species called the Red List.

WILKINSON MICROWAVE ANISOTROPY PROBE, WMAP (MICROWAVE ANISOTROPY PROBE, MAP): A satellite launched in 2001 from Florida, U.S.A. to study microwave background radiation in the universe. In 2003, a full-sky picture was obtained of the early universe showing temperature fluctuations at high resolution. The results supported the big-bang and inflation theories and gave an age for the universe of 13.7×10^9 years.

WILL-O'-THE-WISP (IGNIS FATUUS): Balls of light, usually of bluish colour that are sometimes seen over marshy ground at night. It is believed to be caused by spontaneous combustion of gas containing methane that develops from the breakdown of organic matter in persistently wet areas.

WILLIAMSON'S SYNTHESIS: Either of two methods of producing ethers, both named after the British chemist Alexander Williamson (1824-1904). The methods are: (i) the synthesis of ethers utilising an alkyl iodide and sodium alcoholate or a method of synthesising ethers from alkyl iodides and alkoxides; (ii) the dehydration of alcohols using concentrated sulfuric acid. The reaction can be written as: $2ROH \rightarrow H_2O + ROR$. The method is used for making ethoxythane ($C_2H_5OC_2H_5$) from ethanol by heating at $140^\circ C$ ($413 K$) with excess alcohol (excess acid at $170^\circ C$, $443 K$ gives ethane).

WILSON CLOUD CHAMBER: Apparatus developed to study how water vapour and light interact. It is a cloud chamber containing air supersaturated with water vapour by sudden expansion, in which rapidly moving nuclear particles such as alpha or beta rays produced ionisation tracks by condensation of vapour on the ions produced by the rays.

WILSON'S THEOREM: In mathematics, it is the result that a natural number n is prime if and only if n divides $1 + (n - 1)!$.

WILSON, CHARLES THOMAS REES: Scottish Nobel laureate and physicist (1869-1959), who invented the cloud chamber, which gave the first pictures of the paths of subatomic particles and became an essential tool in the fields of meteorological and atomic physics.

WILT: 1. In botany, of a plant, leaf or flower, to become limp or flaccid, the plant cells lose rigidity (turgor). It is caused by lack of water, too much heat or disease, for example, *fusarium wilt*. 2. A virus disease of various caterpillars, characterised by the liquefaction of body tissues.

WILTING POINT (PERMANENT WILTING POINT): The minimal point of soil moisture a plant requires not to wilt. At this point, any decrease in soil moisture will result in wilting: the plant's leaves dry out, droop, and wither. This is determined by **soil water holding capacity**, which is the amount of water volume between field capacity (the maximum amount of water the soil can hold) and wilting point (where the plant can no longer extract water from the soil). This point of soil water content is generally accepted as about 15 bars tension. At this point, the water film on soil particles is so thin that the energy exerted by plants fails to extract water in sufficient quantities to prevent wilting. If moisture decreases to this or any lower point a plant wilts and can no longer recover its turgidity when placed in a saturated atmosphere for 12 hours.

WIMP: 1. Also known as **weakly interacting massive particle**, it is a hypothetical elementary particle that might make up most of the matter in the universe and that is also predicted to exist in supersymmetry theory, serving as one possible solution to

the dark matter problem. The nature of the dark matter is not yet known, although many experiments are under way to try to discover it directly or indirectly. These particles interact through the weak force and gravity and possibly through other interactions no stronger than the weak force. Because they do not interact with electromagnetism they cannot be seen directly and because they do not interact with the strong nuclear force they do not react strongly with atomic nuclei. 2. In computing, it is acronym for **windows, icons, menus, pointer**, a term used to indicate graphical user interfaces (GUI) of personal computers in operating systems such as Windows, Macintosh and Linux.

WIMSHURST MACHINE: A laboratory electrostatic generator developed by British inventor, James Wimshurst (1832-1903), consisting of two insulating discs to which radial strips of metal foil are attached. After a few strips have been charged individually, the discs are rotated in opposite directions and the charge produced on the strips by induction is collected by metal combs or brushes. These machines belong to a class of generators called influence machines, which separate electric charges through electrostatic induction or influence.

WIN32: The Windows application programming interface (API) for developing 32-bit applications. It has been used for Windows 95, Windows 98, Windows NT and newer Windows operating systems.

WINCH: A device for facilitating the hoisting or hauling of loads. It comprises a rotatable drum around which is wound a rope or cable attached to the load. The drum is turned by means of a hand-operated crank or a motor.

WINCHESTER DISK: Removable disk drive from IBM that put the heads and platters (disks) in a sealed unit for greater speed.

WIND: A volume of gases on a large scale moving over the Earth surface from high pressure areas (anticyclones) to low pressure areas (depression). The large scale patterns of wind and pressure result mainly from differences in the net radiation received at the Earth's surface and modified by rotation of the Earth, mountains and other barriers, relative distribution of land and water bodies and the positions of ocean current.

WIND PATTERNS: Air movement caused by temperature change and influenced by physical features such as mountain ranges, semi-permanent pressure systems, and the shape and rotation of Earth. These winds are often grouped together as trade winds, easterlies, and westerlies.

WIND POLLINATION (ANEMOPHILY): A form of pollination in which pollen evolved by plants such as rice, corn, some grasses and pines for accomplishing cross-pollination are carried by the wind. Wind pollination or anemophilous plants may be either gymnosperms (non-flowering) or angiosperms (flower-producing).

WIND POWER (WIND ENERGY): The extraction of kinetic energy from the wind and conversion of it into a useful type of energy such as thermal, mechanical or electrical. When harnessed, wind energy can be converted into mechanical energy for performing work such as pumping water, grinding grain and milling lumber. For example, vanes or blades mounted on a windmill converts the kinetic energy of wind to a rotary motion which turns a rotor attached to a generator to produce electricity.

WIND SHEAR: The amount of change in the wind's direction or speed over a relatively short distance in the atmosphere with increasing altitude; caused by thunderstorms. There are two types of wind shear in the atmosphere: vertical or horizontal. Vertical wind shear travels at different heights or in a vertical direction and can influence strong thunderstorms. Horizontal wind shear occurs over a horizontal distance, usually influencing directional changes. An aircraft might encounter horizontal wind shear when flying near mountainous areas or when passing directly through a front.

WIND TUNNEL: A chamber through which air is forced at controlled velocities in order to study the effects of aerodynamic flow around airfoils, scale models or other objects.

WIND TURBINE: A machine that converts the wind's kinetic energy into rotary mechanical energy, which is then used to do work. In more advanced models, the rotational energy is converted into electricity by a generator. Wind turbines can be either large propeller type motors on a tall tower or flexible metal strips fixed to a vertical axle at the top and bottom.

WIND-CHILL FACTOR: The temperature of windless air that would have the same effect on exposed human skin as a given combination of wind speed and air temperature.

WINDBREAK: In agriculture, rows of trees planted between fields of crops to prevent the wind from causing erosion of the soil.

WINDPIPE (TRACHEA): The tube-like organ in air-breathing vertebrates leading from the larynx and carrying air to the lungs, through the bronchi, bronchioles and alveoli. It is a tube supported by incomplete rings of cartilage that conducts air from the throat to the bronchi.

WINDING NUMBER/INDEX: The number of times, a closed curve in a plane passes around a given point in the counter clockwise direction. The winding number depends on the orientation of the curve and is negative if the curve travels around the point clockwise. Winding numbers are fundamental objects of study in algebraic topology and play an important role in vector calculus, complex analysis, geometric topology, differential geometry and physics.

WINDMILL: A machine which converts the energy of wind into rotational motion by means of adjustable vanes called **sails** to drive machinery, for example, to generate electricity, grind corn or pump water.

WINDWARD: The side facing the prevailing wind and therefore wetter than its leeward side (side facing away from the wind).

WINDOW: 1. An opening constructed in a wall or roof that functions to admit light or air to an enclosure and is often framed and spanned with glass, mounted to permit opening and closing. 2. A band of electromagnetic wavelengths that is able to pass through a particular medium with little reflection or absorption. An example is a radio window in the atmosphere allowing radio waves of wavelengths 5 mm to 30 mm to pass through. This window enables radio telescopes to be used on the surface of the Earth. 3. In computers, it refers to an area of a computer monitor, usually rectangular that presents its contents such as the contents of a directory, a text file or an image seemingly independently of the rest of the screen. Modern computers, such as the Macintosh and most personal computers, divide the screen into discrete sections, known as windows, within which different pieces of software can be run at the same time - although not necessarily strictly at the same time, as normally only one window is active. The user can control the size, shape and positioning of each window. Data such as a piece of text, a picture or numerical data, can be moved or copied and pasted from one window to another. 4. A period of time during which an activity can or must take place in order to achieve a desired result. An example is a launch window, a period during which a space vehicle must be launched to achieve a planned encounter. 5. An area at the outer limits of the Earth's atmosphere through which a spacecraft must pass in order to return safely. 6. A rectangular area on the screen that displays its own file or message independently of the other areas of the screen. 7. Also called **chaff**, it is strips of foil dropped from an aircraft to confuse enemy radar.

WINDOWS: The name of a range of several different Graphical User Interface (GUI) operating systems produced by the Microsoft Corporation. Windows 3.0 and Windows 3.1 were the first operating systems of this type, produced by Microsoft, to appear in the early 1990s. Microsoft Windows is currently the most widely used GUI for personal computers. It exists in various versions, such as Windows 95, 98, ME, NT, 2000 and XP.

WINDOWS 7: A version of Microsoft Windows operating system released by Microsoft on October 22, 2009 as the successor to Windows Vista (which itself had followed Windows XP). Enhancements

and new features in Windows 7 include multi-touch support, Internet Explorer 8, improved performance and start-up time, Aero Snap, Aero Shake, support for virtual hard disks, a new and improved Windows Media Centre and improved security. Another important change in Windows 7 is the replacement of Windows Vista's Quick Launch toolbar with a new "Superbar" that makes it possible for applications to be pinned to the taskbar.

WINDOWS 8: A version of Microsoft Windows operating system released by Microsoft on October 26, 2012, which is one of the biggest changes Microsoft has made to their operating system since Windows 95. It features some new programming and technology to allow it to run faster and provide a better look and feel to users. It is not only available for desktop and laptop computers, but it is also available to some tablet computers. Windows 8 includes a tablet interface, called Metro, which is compatible with touch-screen displays, like those on a tablet computer. Users can still use the standard windows desktop interface as well, if they prefer. The new Windows 8 Start screen, replaces the start button.

WINDOWS EXPLORER (FILE EXPLORER; EXPLORER): A file browser found in Microsoft Windows 95 and above to explore and manage the drives, folders and files on the computer. Although similar, Microsoft Windows Explorer is not the same as "My Computer" and is not "Microsoft Internet Explorer", even though Internet Explorer is sometimes improperly referred to as Explorer. Windows Explorer can be accessed by clicking the "Start" menu, right-clicking "Computer" and selecting "Explore."

WINDOWS INTERNET NAME SERVICE (WINS): A service that enables Windows to identify NetBIOS systems on a TCP/IP network. It maps NetBIOS names to IP addresses, which is a more standard way to identify network devices.

WINDOWS OPEN SERVICES ARCHITECTURE (WOSA): In computer science, it is a set of interfaces defined by Microsoft to enable Windows programs from different vendors to communicate with one another.

WINDOWS RT (WinRT): A version of Microsoft Windows operating system released by Microsoft on October 26, 2012, same day as Windows 8 and the Microsoft Surface tablet for mobile PCs with ARM-based CPUs. The Windows RT user interface is identical to Windows 8 and includes the same tile-based Start screen. It also provides access to the traditional Windows desktop, where users can manage files and folders and run applications. It can use a standard keyboard and mouse, but it is designed primarily for touchscreen input. The operating system features an "always on" design, which means it receives mail, contacts and calendar updates even while in standby mode.

WINDOWS SOCKET (WINSOCK): In computer science, it is a program that supplies an interface between Windows software in Internet applications.

WINDOWS VISTA: Microsoft's Windows operating system, which is an upgrade to Microsoft Windows XP and Windows 2000. The business version was released at the end of 2006, while the consumer version was released on January 30, 2007. The Vista operating system includes an updated look from Windows XP, referred to as the "Aero" interface. Its features include: (i) Detects hardware problems before they occur; (ii) Windows Vista security features protect against the latest generation of threats, such as worms, viruses and spyware; (iii) Faster start-up time and low power consumption of the new sleep state, etc.

WINDOWS XP (WINDOWS eXPERIENCE): Microsoft's Windows operating system which was introduced in 2001 and was the most significant upgrade to the Windows operating system since Windows 95. The previous version of Windows is Windows Me (or Millennium Edition). Windows XP comes in two versions, Home and Professional.

WINDOWS-BASED TERMINAL (WBT): Network computer that display Microsoft Windows programs, such as Word and Excel, from a server computer because they do not have a hard drive and little

or no installed software but has access to software that is managed and delivered by network servers. A thin client software is used to input keyboard and mouse to the server, which then sends back a screen display.

WINDOWS, ICONS, MENUS, POINTER (WIMP): A term used to indicate graphical user interfaces (GUI) of personal computers in operating systems such as Windows, Macintosh and Linux.

WINE: In computer science, it is a software program that enables users to run many Microsoft Windows applications under Linux.

WING: 1. The modified forelimb of birds and bats or membranous outgrowths of the exoskeleton of insects, used for flight. 2. Either of the two narrow lateral petals of a 'pea' flower. 3. The membranous outgrowth of certain fruits, for example, the samara. 4. A flange running down a stem or stalk as, for example, seen along the stems of the hairy vetchling (*Lathyrus hirsutus*).

WING COVERT: A feather that covers the base of a main wing feather, forming a number of rows. In flight, the coverts help to smoothen airflow over the wings and tail.

WING-BAR: A line of colour, usually contrasting across the middle, tip, or base of a bird's wing, which has been made by the wing coverts.

WING-FLICKING: Rapid movement made by a bird's wings when it is not in flight and is at rest.

WINKLER TITRATION: A chemical method for estimating the dissolved oxygen in seawater.

WINNOW: 1. To separate grain from its husk chaff by blowing air through it or tossing it in the air. 2. A privacy technique that is simpler than encrypting a message with a key. For example, chaff is the unusable part of a plant and winnow means to separate the chaff so it can be discarded. Thus, chaff is sent (fake packets) along with real packets and the chaff is winnowed at the other end by comparing a message authentication code (MAC) stored in each packet. The receiving end knows the valid MACs and those which are not.

WINO: One of the fermion partners such as gluino, photino and zino of existing bosons. The boson partners of existing fermions have names like selectron, squark and slepton. In the supersymmetry theories, every boson has a corresponding fermion partner and every fermion has a corresponding boson partner.

WINOGRADSKY COLUMN: A glass column that allows growth of microorganisms by simulating conditions similar to those found in nutrient-rich water and sediments. This column contains an anaerobic lower zone and an aerobic upper zone. The cylinder is incubated in subdued sunlight for several weeks, and a mixture of organisms develops. It was named after the Russian microbiologist, Sergei Winogradsky, who designed it to study soil microorganisms.

WINSOCK (WINDOWS SOCKET): In computer science, it is a program that supplies an interface between Windows software in Internet applications.

WINTER: The coldest season of the year in temperate regions, occurring between autumn and spring, extending in the Northern Hemisphere from the winter solstice to the vernal equinox. It occurs from November/December to February/March in the Northern hemisphere and in the Southern hemisphere from June to August. In some places in temperate regions, temperatures can drop below freezing point.

WINTER ANNUAL: A plant such as henbit, deadnettle, chickweed and winter cress, grown outdoors and germinates in autumn or winter, when the soil temperature is cool, live through the winter, then bloom in winter or spring the following year and dies.

WINTER SOLSTICE: The shortest day and the longest night of the year, which occurs about 22 December in the Northern Hemisphere and 21 June in the Southern Hemisphere. In the Northern Hemisphere, the Sun is at its farthest point in the south from the Earth's equator. In the Southern Hemisphere, the Sun is at its farthest point in the north from the Earth's equator. The number of hours of daylight depends on the latitude of a location. For example, in London on 1st December 2017, there was 8 hrs 10mins 54 sec of daylight; on the same day in

Rekjavik in Iceland, there was 5 hrs 0 mins 36 sec of daylight.

WINTER BUD: A hibernating vegetative shoot.

WIRE: A long flexible strand of metal used to bear mechanical loads and to carry electricity and telecommunications signals. It is commonly formed by drawing the metal through a hole in a die or draw plate. Standard sizes are determined by various wire gauges.

WIRE GAUZE: A very fine, woven wire that looks like netting, often used in science laboratories as a flame shield, for example, it may be attached to the ring stand between a Bunsen burner and a beaker or a flask during heating to provide uniform heating.

WIRED EQUIVALENT PRIVACY (WEP): A wireless security protocol, used for wireless local area network (WLAN) and defined in the IEEE Wireless Fidelity (Wi-Fi) 802.11b standard.

WIRELESS: The original name for radio receivers or a term describing the transfer of information over a distance by radio signals rather than wires or communication systems and devices that make use of cell phone technology. It encompasses various types of fixed, mobile and portable two-way radios, cellular telephones, personal digital assistants (PDAs) and wireless networking, GPS units, wireless computer mice, keyboards and headsets, satellite television and cordless telephones. The distances involved may be short (a few metres as in television remote control) or long (thousands or millions of kilometres for radio communications).

WIRELESS APPLICATION PROTOCOL (WAP): A specification for a set of communication protocols that enables a user to browse online services such as e-mail, chat, information relating to the weather, traffic conditions and shopping via a special type of mobile phone. It was started in the 1990s by Ericsson, Motorola, Nokia and Unwired Planet (now known as Openwave) in June 1997 and now includes hundreds of member companies that are infrastructure providers, software companies and content providers. WAP is the mobile phone equivalent of the World Wide Web. Newer mobile phones include WAP browser software to allow users access to WAP sites.

WIRELESS EQUIVALENT PRIVACY (WEP): An encryption protocol defined in the 802.11 standard that provides a wireless local area network (WLAN) with a level of security and privacy. It was later replaced by WPA, which is an improved encryption standard introduced with 802.11i.

WIRELESS FIDELITY (WIFI): A term used to refer to a networking technology that allows computers and other devices to communicate without a Local Area Network (LAN) that uses cable connections.

WISWESSER LINE NOTATION (WLN): An early line notation for chemical structures, invented by William J. Wiswesser in 1949. The symbols used are the upper-case letters of the alphabet (A-Z), the numerals (0-9), with three other symbols: the ampersand (&), the hyphen (-) and the oblique stroke (/) and a blank space. It was the basis of ICI Ltd's CROSSBOW database system developed in the late 1960's. It was also the tool used to develop the CAOICI (Commercially Available Organic Chemical Intermediates) database, the data file from which MDL's ACD file was developed. WLN is still being extensively used by BARK Information Services.

WITHIN-SUBJECTS DESIGN (REPEATED MEASURES DESIGN): In statistics, it is a type of an experiment, which repeatedly measures the values of the dependent variable for the same subjects under the various experimental conditions and the values are compared.

WIZARD: In computer terminology, it refers to a software that guides the user step-by-step through a complex task, such as setting up a software on a network or configuring a printer to output data in a special format. For example, a wizard in Microsoft Word can help create and format a new document according to your needs.

WLAN (WIRELESS LOCAL AREA NETWORK): A wireless computer network that links two or more devices that use high-frequency radio waves rather than wires to transmit data within a limited area such as a home, school, computer laboratory, or office building. This gives users the ability to move around within a local coverage area and yet still be connected to the network. Every component that connects to a WLAN

is considered a station and falls into one of two categories: access points (APs) and clients. APs transmit and receive radio frequency signals with devices able to receive transmitted signals; they normally function as routers. Clients may include a variety of devices such as desktop computers, workstations, laptop computers, IP phones and other cell phones and Smartphones. All stations able to communicate with each other are called basic service sets (BSSs), of which there are two types: independent and infrastructure. Independent BSSs (IBSS) exist when two clients communicate without using APs, but cannot connect to any other BSS. Such WLANs are called a peer-to-peer or an ad-hoc WLANs. The second BSS is called an infrastructure BSS. It may communicate with other stations but only in other BSSs and it must use APs.

WMAP (WILKINSON MICROWAVE ANISOTROPY PROBE; MICROWAVE ANISOTROPY PROBE, MAP): A satellite launched in 2001 from Florida, U.S.A. to study microwave background radiation in the universe. In 2003, a full-sky picture was obtained of the early universe showing temperature fluctuations at high resolution. The results supported the big-bang and inflation theories and gave an age for the universe of 13.7×10^9 years.

WNT PROTEIN: Any of a family of secreted signalling molecules crucial in regulating cell proliferation, differentiation and survival both during embryological development and in later life. When misregulated, it can contribute to several types of cancer.

WOBBLE: A proposed explanation for base-pairing that is non-Watson-Crick type, which often occurs between the 3' base in the codon and the 5' base in the anticodon (a set of 3 nucleotides in transfer RNA involved in the formation of a protein). Wobble base pairs are fundamental in RNA secondary structure and are critical for the proper translation of the genetic code.

WÖHLER'S SYNTHESIS: A method of preparing urea ($\text{CO}(\text{NH}_2)_2$) by heating ammonium isocyanate (NH_4NCO), performed by Friedrich Wöhler in 1828. It was the first synthesis of an organic compound. Before that it was thought to be produced only by living organisms.

WOLF-RAYET STAR: Extremely hot, bright and massive stars, with surface temperatures between 25,000 K to 50,000 K, often found inside a developing planetary nebula, which are losing mass rapidly by means of a very strong stellar wind, with speeds of up to 2000 km/s.

WOLFRAM LAMP: Incandescent lamp that generates light by passing an electric current through a thin filament wire (usually of wolfram) until it is extremely hot. The lamps are often filled with a halogen gas such as iodine and bromine that allow filaments to work at higher temperatures and higher efficiencies.

WOLFRAMITE (IRON MANGANESE TUNGSTEN): A brownish black crystalline mineral, the main ore of tungsten and wolfram made up of a mixture of iron manganese tungstate, $(\text{Fe,Mn})\text{WO}_4$ with Moh's scale hardness of 5 to 5.5 and relative density of 7 to 7.5. It crystallises in the monoclinic system, exhibits perfect cleavage and shines with a submetallic to resinous lustre. It occurs in combination with such minerals as cassiterite, scheelite, quartz, pyrite, galena, sphalerite and arsenopyrite.

WOLLASTON PRISM: A type of quartz prism, invented by William Hyde Wollaston, for producing plane-polarised light. It deviates the ordinary and extraordinary rays in opposite directions by approximately the same amount. It can be used with ultraviolet radiation. It consists of two orthogonal calcite prisms, cemented together on their base to form two right triangle prisms with perpendicular optic axes. Outgoing light beams diverge from the prism, giving two polarised rays, with the angle of divergence determined by the prisms' wedge angle and the wavelength of the light.

WOMB (UTERUS): A flattened, pear-shaped, hollow organ in the pelvis of the human female and most other mammals, where the embryo and foetus of mammals develop before birth. It is connected to the vagina by the cervix. The womb or uterus consists of a body, a base or fundus, a neck or cervix and a mouth. Suspended in the pelvis, it lies with the base directed upward and forward and the cervix directed slightly backward. On either side of the womb or uterus lies

an ovary. Eggs produced by the ovaries reach the uterus through the fallopian tubes.

WOOD: The hard structural and water conducting tissue of many perennial plants. It is the xylem tissue that forms the bulk of the stem of woody plant and shrubs. The wood of gymnosperms such as pine and fir is called **softwood**. The wood of angiosperms such as oak, odum and mahogany is called **hardwood**.

WOOD ALCOHOL (METHYL ALCOHOL; METHANOL; CARBINOL): Colourless, toxic, flammable liquid with the chemical formula CH_3OH , melts at -97.8°C (175 K), boils at 64.7°C (337.7 K) and has a relative density of 0.7915 at 20°C (293 K). It is a kind of alcohol distilled from wood. It is now synthesised from hydrogen and carbon monoxide. It is used as fuel, in the synthesis of many organic compounds, particularly formaldehyde, as a solvent and a denaturant for ethyl alcohol.

WOOD HORN: A mechanical device that acts by absorption as a perfect photon trap.

WOOD LAMP: A term used to describe a low-pressure mercury arc possessing a fluorescing layer, which emits in the UV-A region (from 315 nm to 400 nm).

WOOD LOUSE (PILLBUG; SLATER; SOWBUG): Any of various terrestrial isopod crustaceans of the suborder Oniscidea, having a grey or brown oval segmented body and fourteen jointed limbs. It is commonly found in damp places such as under logs. Some woodlice can roll into a ball as a defensive mechanism.

WOOD PITCH: The sticky residue left after wood tar has been distilled or the condensed liquid produced from burning charcoal gases. It is used as a binder in briquettes.

WOOD PULP: A pulpy mass of fibres processed from wood, used for making paper, rayon and other cellulose fibres and plastics.

WOOD SUGAR (XYLOSE): An aldopentose sugar, widely found in plants, particularly in woody tissues. It occurs mainly in its polymerised form, xylan. It is also a constituent of the rare disaccharide primeverose.

WOOD TAR: A black, viscous fluid, a by-product in the destructive distillation of wood. It is used as a protective coating for rope and timber and also used in pitch, preservatives and medicines.

WOOD'S METAL (LIPOWITZ'S ALLOY; CERROBEND; BENDALLOY; PEWTALLOY): An alloy with a low melting point 71°C (344 K) used for fusible links in automatic sprinkler systems. It is made of bismuth (50%), lead (25%), tin (12.5%) and cadmium (12.5%). Varying the composition changes the melting point.

WOODWARD-HOFFMANN RULES (FRONTIER-ORBITAL THEORY): Rules governing the formation of products during certain types of organic concerted reactions, named after R. B. Woodward and Roald Hoffmann, the developers. The theory is also concerned with the way the orbitals of the reactants change continuously into orbitals of the products during reaction and with conservation of orbital symmetry during the process.

WOOFER: A large loudspeaker used to reproduce audio signals of relatively low frequency in conjunction with a tweeter. It is often a mid-range speaker, in a high-fidelity sound reproducing system.

WOOL: The protective covering of sheep, goats and similar animals and a natural protein animal fibre which has a crinkled and scaly structure. The fibres are elastic and can trap air and warmth. It is processed and used for clothing, blankets, carpets, felt, wool insulation, upholstery, etc.

WORD: 1. Amount of data, whose length, called the **word-length** usually constitute a group of bits (binary digits) which are often 32, 48 and 64 that a computer's central processing unit treats as a single working unit. It is the basic unit of storage on a digital computer. In general, increasing the word length leads to a faster and more powerful computer. 2. **Microsoft word** is a full-featured word processing program for Windows and the Macintosh from Microsoft. Included in the Microsoft application suite, it is a sophisticated program

with rudimentary desktop publishing capabilities that has become the most widely used word processing application on the market. The first versions of Word came out under Disk Operating System (DOS) and provided both graphics-based and text-based interfaces for working with a document. 3. In mathematics, it is an expression of the form $x_1^{\pm 1} x_2^{\pm 1} \dots x_n^{\pm 1}$. With the product of such non-empty words defined as concatenation, they form a semi-group and when this is extended to empty words, 1, by $1u = u$ for all words u , the set of all words is a monoid.

WORD BLINDNESS (DYSLEXIA): A malfunction in the brain's synthesis and interpretation of written information, characterised by difficulties in acquiring and processing language that is typically manifested by a lack of proficiency in reading, spelling and writing, although the person may excel in other areas, for example, in mathematics. A similar disability with figures is called **dyscalculia**. Acquired dyslexia may occur as a result of brain injury or disease.

WORD PROCESSOR: A computer program that allows the input, amendment, manipulation, storage and retrieval of text upon command or a computer system that runs such software.

WORK: A measure of the amount of energy transferred by a force acting through a distance or the transfer of energy, measured as the product of the force applied to a body and the perpendicular distance moved by that body in the direction of the force. Work is the scalar product to the force and displacement vectors. If the force and the displacement are parallel and in the same direction, the mechanical work is positive; if the force and the displacement are parallel but in opposite directions, the mechanical work is negative. If the force and the displacement act perpendicular to each other, zero work is done by the force. The SI unit of work is the joule (J), which is defined as the work done by a force of one newton acting over a distance of one metre. Its symbol is W . Work (W) = force (F) $\times$ distance of displacement in the direction of the force (S): $W = FScos\theta$.

WORK DONE: The work done by a force acting on a body is the product of the force and the perpendicular distance moved by its point of application in the direction of the force. It can also be defined as the transference of energy that occurs when a force is applied to a body that is moving in such a way that the force has a component in the direction of the body's motion. It is equal to the line integral of the force over the path taken by the body. If a force F acts in such a way that the displacement S is in a direction that forms an angle θ with the direction of the force, then the work done is given by $W = FScos\theta$. The unit of measurement is the joules.

WORK FUNCTION: The quantity with the dimensions of energy that determines the extent to which thermionic or photoelectric emission of a solid will occur at a given temperature in accordance with Einstein's photoelectric equation or the Richardson equation. It is either expressed as the minimum energy needed to remove an electron in electron volts or joules (symbol W) and called the work function energy; or as a potential difference (symbol e) in volts called the work function potential.

WORK HARDENING (STRAIN HARDENING; COLD WORKING): Strengthening of a metal by plastic deformation or by working them cold. This strengthening occurs because of dislocation movements within the crystal structure of the material, which causes a permanent distortion of the crystal structure. Examples of metals that can be work hardened are iron, copper, aluminium, etc; and those that cannot be work hardened are lead and zinc, because they can form crystals again at room temperature.

WORK SOFTENING: The application of a finite shear to a system after a long rest may result in a decrease of the viscosity or the consistency. If the decrease persists when the shear is discontinued, this behaviour is called work softening (or shear breakdown), whereas if the original viscosity or consistency is recovered this behaviour is called **thixotropy**.

WORKBENCH: A strong table of various sizes on which objects such as pieces of wood or metal can be worked upon using tools. A

laboratory bench is a type of workbench.

WORKER: A member of a colony of social insects such as ants, bees, wasps or termites, usually a sterile female but often a sexually immature individual of either sex, that performs specialised work such as building the nest, collecting and storing food and feeding other members of the colony.

WORKING ELECTRODE: An electrode that serves as a transducer responding to the excitation signal and the concentration of the substance of interest in the solution being investigated and that permits the flow of current sufficiently large to effect appreciable changes of bulk composition within the ordinary duration of a measurement.

WORKSHEET (SPREADSHEET): A data file made up of rows and columns that are used to sort data and allow a user to manipulate and arrange data easily, commonly numerical data. What makes a spreadsheet software program most unique is its ability to calculate values using mathematical formulas and the data in the cells. Example of a worksheet is Microsoft Excel spreadsheet.

WORKSTATION: A powerful personal computer used for business or professional work such as engineering applications (CAD/CAM), desktop publishing, software development and other types of applications. It has one or more high resolution displays and is faster than a personal computer. It is commonly connected to a local area network and run multi-user operating systems.

WORLD CONSERVATION UNION (INTERNATIONAL UNION FOR THE CONSERVATION OF NATURE, IUCN): Founded in 1948, it is an international body that is concerned with finding pragmatic solutions to the most pressing environment and development challenges in the world and encourages the preservation of wildlife, natural environments and living resources. It has a membership of about 1,300 government and NGO members and more than 15,000 volunteer experts in 185 countries. Its activities include data gathering and analysis, research, field projects, advocacy, lobbying and education.

WORLD LINE: The unique path of a particle as it travels through 4-dimensional spacetime. The position of a particle at time t can be found by slicing space-time at time t and finding where the slice cuts the world line of the particle.

WORLD WIDE WEB (WWW; THE WEB): A hypermedia computer-based information service system, distributed over a large number of computer sites that allows users to view web pages that may contain text, images, videos and other multimedia and navigate between them by using hyperlinks and retrieve information from documents containing links to the internet. It was developed in the early 1990s by CERN (Conseil Européen Pour la Recherche Nucléaire), now known as European Laboratory for Particle Physics, an European Union organisation that carries out research into high energy particle physics.

WORLD WIDE WEB CONSORTIUM (W3C): A non-profit international organisation that work together to develop standards for the World Wide Web and also develop relevant protocols and guidelines. The organisation was founded by Tim Bernes-Lee (1955 -), British physicist in 1994, who is also credited with the following: (i) invention of the World Wide Web (WWW); (ii) proposed hypertext; and (ii) created the first website.

WORM: 1. An elongated limbless invertebrate animal with a soft body with no appendages. The major phyla are Platyhelminthes (flatworms), Annelida (annelids or segmented worms), Nemertea (ribbon worms), Acanthocephala (spiny-headed worms) and Aschelminthes (nematodes and others). 2. A computer program that uses a computer network to send copies of itself to other computers on the network, replicates itself to prevent deletion and interferes with the host computer's operation. Unlike a computer virus, it does not need to attach itself to an existing program. 3. In computer science, it is abbreviation for **write-once, read-many**, a storage device, similar to a CD-ROM, in which data is written to a disc once using a low-powered laser that makes permanent marks on the surface thus

preventing the drive from erasing the data or overwriting the same area. In computer science, it is a storage device, similar to a CD-ROM, on which data is written once using a low-powered laser that makes permanent marks on the surface thus preventing the drive from erasing the data or overwriting the same area. The data can be read any number of times. WORMs are mainly used for archiving and back-up copies. 4. A computer program that uses a computer network to send copies of itself to other computers on the network, replicates itself to prevent deletion and interferes with the host computer's operation. Unlike a computer virus, it does not need to attach itself to an existing program.

WORM-LIKE CHAIN (CONTINUOUSLY CURVED CHAIN): A hypothetical linear macromolecule consisting of an infinitely thin chain of continuous curvature. The direction of curvature at any point is random. The model describes the whole spectrum of chains with different degrees of chain stiffness from rigid rods to random coils and is particularly useful for representing stiff chains. This chain is sometimes referred to as a **Porod-Kratky chain**.

WORMHOLE: A theoretical structure in a region of the universe space-time, bridging two universes through a path that is shorter in distance or duration than would otherwise be expected.

WORONIN BODY: A membrane-bound proteinaceous spherical body occurring in the hyphae of members of the Ascomycotina, named after the Russian botanist Mikhail. Each perforated septum of the hypha has associated woronin bodies located on the other side of the septum. When occasion demands the closure of the pore, hydrostatic pressure of the cytoplasm pushes the woronin body towards the pore to plug it.

WOSA (WINDOWS OPEN SERVICES ARCHITECTURE): In computer science, it is a set of interfaces defined by Microsoft to enable Windows programs from different vendors to communicate with one another.

WOUND CORK: A layer of phellem that forms over a damaged part of the plant. It prevents desiccation of the underlying tissues and the entry of pathogens.

WOUND HORMONE: Any substance produced by damaged cells that diffuses into nearby undamaged cells and there stimulates meristematic activity, resulting in the formation of a protective callus. Auxins, gibberellins and certain products of wounding have been implicated in the wounding reaction.

WRENCH: Any manually operated or power tool for gripping and turning or twisting the head of a bolt, a nut, a pipe, etc. It usually consists of fixed or adjustable jaws with a handle. A wrench is actuated by a force applied at one end only and, in addition to the turning moment, it creates an unbalanced force on the element being tightened.

WRITE: To transfer data to a storage medium such as a magnetic or optical disk or tape or to display images or text on a computer monitor. Every write operation in the computer implies a read operation from some other place. For example, to write a file to a disk requires reading it from somewhere else. Write is analogous to record on an audio recorder or videotape recorder.

WRITE ONCE, READ MANY (WORM): In computer science, it is a storage device, similar to a CD-ROM, on which data is written once using a low-powered laser that makes permanent marks on the surface thus preventing the drive from erasing the data or overwriting the same area. The data can be read any number of times. WORMs are mainly used for archiving and back-up copies.

WRITE PROTECTION (WRITE PROTECT): A device on a storage medium such as a computer disk and tape that provides data security by allowing data to be read but not deleted, altered or overwritten. For example, Transcend JetFlash USB flash drives have small switches, for write protection and Secure Digital (SD) cards have a write-protect tab on the left side. This protection may be deactivated by covering the tap with a self-sticking tape and reactivated by removing the tape.

WRONSKIAN: A determinant used in the study of differential equations, where it can sometimes show linear independence in a set of solutions. For n functions on an open interval, it is the determinant of the matrix of which the i, j th entry is the $(j - 1)$ th derivative of the i th function evaluated at x . Given that the functions have $n - 1$ continuous derivatives on $]a, b[$, the functions are linearly independent if the Wronskian is not identically zero. Conversely, if the Wronskian vanishes at even one point and the n functions solve an n th-order linear differential equation with continuous coefficients of which the leading coefficient is never zero, then the functions are linearly dependent and the Wronskian vanishes everywhere. It is named after Polish philosopher, lawyer, mathematician and physicist, Józef Maria Hoene-Wroński (1776-1853).

WROUGHT IRON: A highly refined iron containing less than 0.1% carbon and 1-3% of slag (mostly iron silicate), widely used for construction work before the days of cheap steel. It is strong, tough and easy to manage. Steel has now supplanted wrought iron for structural purposes.

WURTZ REACTION: A method of synthesising hydrocarbons using alkyl iodides and sodium metal according to the reaction $2\text{CH}_3\text{I} + 2\text{Na} \rightarrow \text{CH}_3\text{CH}_3 + 2\text{NaI}$. A similar reaction using a haloalkane and haloarene is called the **fitting reaction**.

WURTZITE STRUCTURE (WURTZITE CRYSTAL STRUCTURE): A type of ionic crystal named after the mineral wurtzite, in which each ion has coordination number of 4 with the anions arranged in a hexagonal fashion, while the cation are seen to occupy rather a tetrahedral hole. Examples of wurtzite structure compounds are wurtzite (AgI), zinc oxide (ZnO), zinc sulfide (ZnS), cadmium sulfate (CdS), cadmium selenide (CdSe), silicon carbide (α -SiC), gallium nitride (GaN), aluminium nitride (AlN) and other semiconductors. Wurtzite crystals generally have properties such as piezoelectricity and pyroelectricity, which centrosymmetric crystals lack.

WVGA (WIDE VGA; WIDE VIDEO GRAPHICS ARRAY): Any display monitor with a height of 480 pixel (same as that of (VGA) and a width of 800. It is commonly used in modern mobile phones.

WYSIWYG (WHAT YOU SEE IS WHAT YOU GET): In computer science, it is an application that enables one to see on the display screen exactly what will appear when the document is printed.

WXGA (WIDESCREEN EXTENDED GRAPHICS ARRAY): Any display monitor with a resolution of 1280 (width) x 800 (height).

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x



x: 1. In Roman notation, it is the symbol for the numeral 10. It is also written in upper case. 2. Often written in upper case, in electricity, it is the symbol for reactance, defined as the opposition to the flow of alternating current caused by the inductance and capacitance in a circuit rather than by resistance. 3. In computer science, it is a wild card character often used to describe versions of hardware or software. One might, therefore, refer to Windows 3 x (any version of Windows, from 3.0 to 3.31) or an x86 chip (any of the chips manufactured by Intel with serial numbers ending in 86).

X.25: In telecommunications, it is a protocol for sending particles of data over a network.

X.400: In computer science, it is one of the set of standards in the 400-number range maintained by the ITU (International Telecommunications Union) that forms the basis for sending global e-mail messages over telecommunication lines.

X.500: An ISO (International Organization for Standardization) and ITU (International Telecommunications Union) global directory standard for managing global directories about objects such as countries, cities, organisations, people, etc., by looking up information by name (a white-pages service) and to browse and search for information (a yellow-pages service). It supports X.400 systems.

X INACTIVATION (X-CHROMOSOME INACTIVATION; LYONIZATION): The inactivation of all but one of the X chromosomes in the cells of female mammals to prevent having twice as many X chromosome gene products as the male, which only possesses a single copy of the X chromosome. In mammals, males receive one copy of the X chromosome while females receive two copies. To prevent female cells from having twice as many gene products from the X chromosomes as males, one copy of the X chromosome in each female cell is inactivated. In placental mammals, the choice of which X chromosome is inactivated is random, whereas in marsupials it is always the paternal copy that is inactivated.

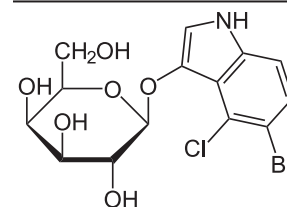
X-AXIS: 1. The horizontal axis of a two-dimensional Cartesian coordinate system. 2. One of three axes (x -, y -, z -) in a three-dimensional Cartesian coordinate system.

X-CHROMOSOME: The larger of the two sex chromosomes, which is associated with female characteristics in mammals, occurring paired in the female and single in the male; the smaller being the Y chromosome. Genes carried on the X chromosome produce the phenomenon of sex linkage.

X-COORDINATES: The position of a point in a two- or three-dimensional Cartesian coordinate system. It is equal to the directed distance of a point from the y -axis in a two-dimensional system or from the plane of the y - and z - axes in a three-dimensional system, measured along a line parallel to the x -axis.

X-GAL (BROMO-CHLORO-INDOLYL-GALACTOPYRANOSIDE [BCIG]):

A compound that causes ice-minus bacteria to turn blue. It is used with cloning vectors containing part of the *lacZ* gene to select recombinants in molecular cloning. The peptide produced by the *lacZ* fragment complements the defective β -galactosidase enzyme in host cells.



Chemical Structure of X-gal

X-INTERCEPT: The point at which a line crosses the x -axis on a graph or where the y value equals 0: x -intercept = $(x, 0)$.

X-PENTOMINO: A pentomino, a five-square polyomino, in the shape of the letter X.

X-RADIATION: Radiation resulting from the interaction of high energy particles or photons with matter.

X-RAY: A type of electromagnetic radiation with very high frequency and short wavelength of 100 nanometres to 0.001 nanometre (between gamma rays and ultraviolet light) produced by the deceleration of charged particles or the transitions of electrons in atoms. It can penetrate solids. They are used in medicine to diagnose bone fractures, dental cavities and cancer; to locate foreign objects in the body; and to stop the spread of malignant tumours. In industry, they are used to analyse and detect flaws such as cracks in structures. Although the chances of being harmed from medical X-rays when used with care is very low, some of the health risks posed by X-ray to humans include the following: genetic damage to reproductive cells such as an ovum in a woman or sperm cells in a man; and damage to living cells.

X-RAY ASTRONOMY: The branch of astronomy that studies the properties of X-ray emission from astronomical objects, from stars to galaxies and quasars by rockets, balloons and satellites. The first non-solar X-ray source was detected in a rocket flight in 1962. X-ray astronomy is traditionally divided into two broad bands: soft and hard, depending on the energy of the radiation being studied. Observations in the soft band (below about 10 keV) must be carried out above the atmosphere from rockets or satellites, while hard X-ray observations can be made at high altitudes achievable by balloons.

X-RAY ATTENUATION: The lessening of the intensity of an X-ray beam by being absorbed by a material or by spreading out of the beam.

X-RAY BINARY: A special class of binary stars where one of the stars is a collapsed object such as a white dwarf, neutron star, or black hole that emits X-rays. These pairs of stars produce X-rays if the stars are close enough that material is transferred from the normal star by the gravity of the dense, collapsed star, producing X rays in the process.

X-RAY CRYSTALLOGRAPHY: The use of diffraction patterns produced by scattered x-rays to study the structure of crystals. Structurally, a crystal is a three-dimensional periodic arrangement in space of atoms, groups of atoms or molecules. The method involves directing a beam of x-rays at a crystalline sample and recording the diffracted x-rays on a photographic plate. The basis of x-ray crystallography is x-ray diffraction.

X-RAY DIFFRACTION: The scattering or bending of waves of x-rays by matter with accompanying variation in intensity in different directions due to interference effects or a method of studying the atomic and molecular structure of crystalline substances by using